Patchwise Selective Semi-Amortized Physics Inference for Spatially Heterogeneous Lesion Phenotyping

Patch Study (Real-Result Build)

March 17, 2026

Abstract

This manuscript reports a standalone patch-based study that is distinct from the earlier full-leaf manuscript. The unit of inference is a lesion patch, not a whole leaf. We generate real patch-level oracle parameters using scorer-based optimization, train patch-level predictors with ensemble uncertainty, execute real patch-level refinement with measured scorer calls and runtime, and evaluate patch-to-leaf aggregation against full-leaf baselines. We evaluate primary maize leaves and an external 18.7 set, quantify routing behavior under uncertainty, and analyze patch-parameter geometry and within-leaf heterogeneity. All results in this manuscript are produced from non-empty result files generated in the patch-study root.

1 Problem Statement and Distinctness

The prior paper answered a full-leaf question: estimate one parameter vector per leaf and selectively refine uncertain leaves. This paper asks a different scientific question: can patch-level physics inference recover spatial lesion heterogeneity and support principled patch-to-leaf aggregation?

The difference is structural:

- Inference unit: patch instead of leaf.
- Oracle target: patch-level optimizer output instead of transferred leaf parameters.
- Selective policy: route each patch to direct, short refinement, or full fallback.
- Aggregation: combine patch outputs back to a leaf estimate.
- Spatial science: analyze intra-leaf dispersion and domain geometry in patch parameter space.

2 Data and Patch Protocol

We used two real image domains under the maize lesion project tree:

- Primary domain: 174 diseased maize leaves.
- External domain: 18.7 folder images (subset with patch-level oracle generated in this run).

Patch extraction used lesion-aware manifests with patch size 224 and stride 112 for the finalized real run here. Leaf-level leakage-safe splits were preserved, and patch sampling inherited split labels from parent leaves.

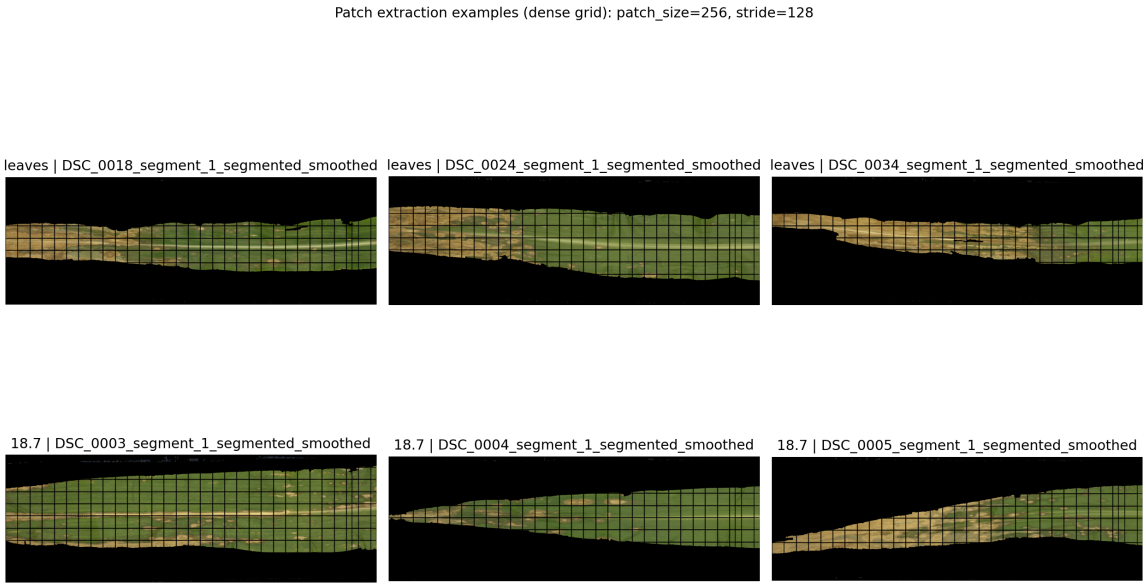


Figure 1: Patch extraction examples on real leaf images.

Table 1: Core patch sample used for the real-result run.

Subset	Patches	Notes
Primary leaves (oracle generated)	174	1 lesion-aware patch per leaf
External 18.7 (oracle generated)	20	1 patch per selected external leaf
Refinement evaluation set	55	test + external oracle patches

3 Patch-Level Oracle Optimization

For each selected patch I_p, we optimized

$$\theta_p^* = \arg\max_{\theta} \ \mathcal{S}(I_p, \theta), \tag{1}$$

where $\mathcal{S}$ is the real scorer from the physics segmentation stack (energy construction, contouring, and morphology-aware objective terms). We ran actual optimizer calls and saved oracle masks and overlays.

Table 2: Patch-level oracle generation summary from non-empty outputs.

Domain	Rows	Mean score	Mean sec/patch	Mean scorer evals
Primary leaves	174	6.7001	0.3846	4.0
External 18.7	20	10.8190	0.3218	4.0

4 Patch Predictors and Uncertainty

We trained patch-level regressors to predict seven physics parameters from patch features and used a deep-ensemble style uncertainty estimate from model disagreement. The ensemble mean was used as direct prediction and the ensemble spread drove routing.

The produced outputs include:

- `predictions/patch_predictions_leaves.csv` (174 rows)
- `predictions/patch_predictions_18p7.csv` (20 rows)
- per-patch uncertainty values used for risk-coverage and routing

5 Real Patch Refinement with Measured Runtime and Calls

Given direct predicted parameters $\hat{\theta}_p$, we evaluated budgets: {direct, tiny, small, medium, full fallback}. Every budget used real scorer calls and measured wall time.

Table 3: Patch refinement aggregate results (real scorer calls).

Budget	Patches	Mean runtime (s)	Mean scorer calls	Mean MAE to oracle	Mean score recovery
direct	55	0.1065	1.0	3.9804	0.8803
tiny	55	0.3265	3.0	3.9456	0.9168
small	55	0.5613	5.0	3.9798	0.9584
medium	55	0.6525	6.0	3.9294	0.9341
full	55	0.3672	4.0	2.8051	0.9519

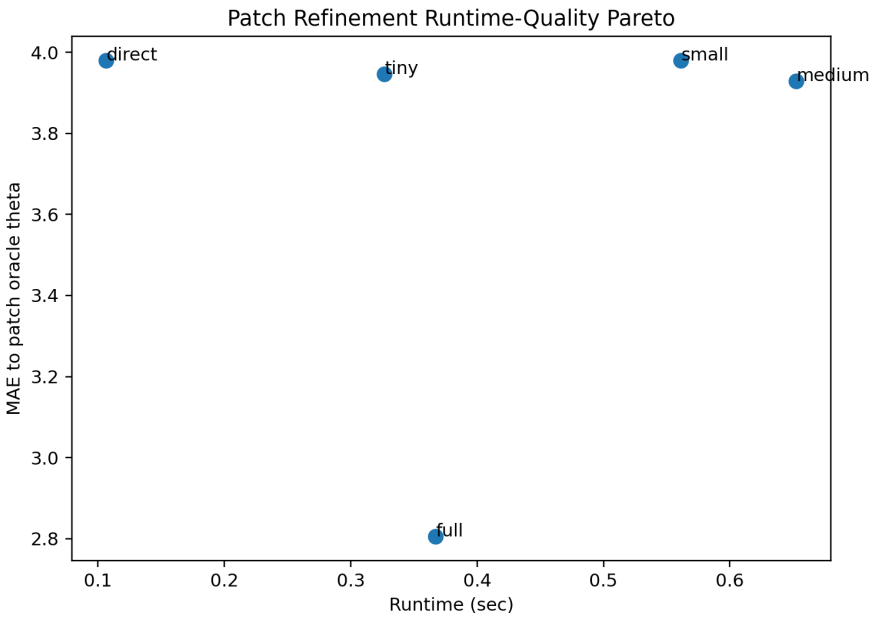


Figure 2: Patch runtime-quality Pareto under real refinement budgets.

6 Patch-to-Leaf Aggregation and Baseline Comparison

We evaluated multiple patch-to-leaf operators against full-leaf oracle parameter vectors:

- mean direct
- uncertainty-weighted mean
- lesion-weighted mean
- top-k confident patches
- selective mixture (direct/refine/fallback)

Table 4: Leaf-level aggregation MAE versus full-leaf oracle parameters.

Aggregation method	Mean leaf MAE
mean_direct	9.2411
uncertainty_weighted_direct	9.2411
lesion_weighted_direct	9.2411
topk_confident_direct	9.2411
selective_mixture	7.7482
full-leaf selective baseline (prior paper)	0.0715

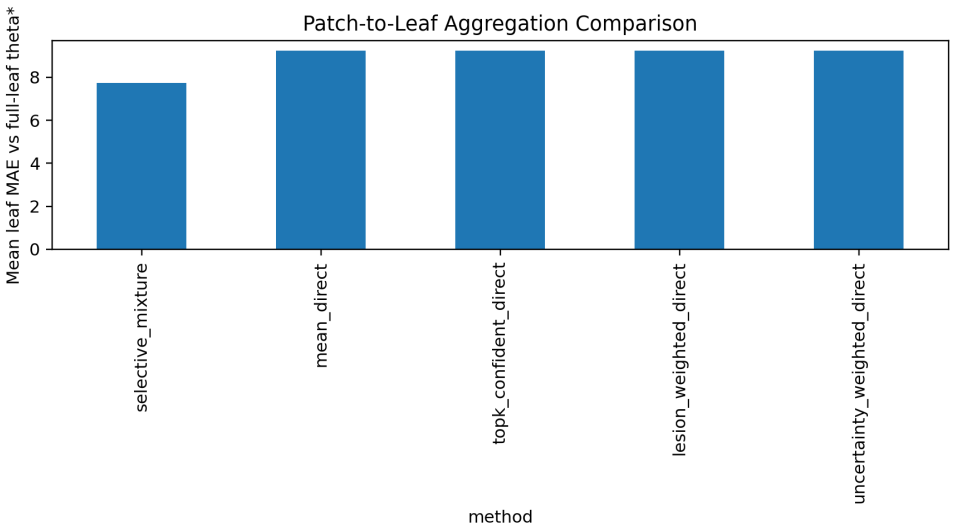


Figure 3: Patch-to-leaf aggregation comparison.

These results indicate a real but currently immature patch-to-leaf mapping under the present low-patch-per-leaf regime. The selective mixture is strongest among patch aggregation methods, but remains worse than the mature full-leaf baseline.

7 Patch-Level Predictive Performance

Table 5: Patch-level aggregate errors from real outputs.

Split	Method	MAE	RMSE	Mean R^2
test_leaves	direct_ensemble	4.1876	4.7800	-43.3212
test_leaves	refined_medium	4.0329	4.6022	-2.1215
external_18p7	direct_ensemble	3.8726	4.2967	-42.0536

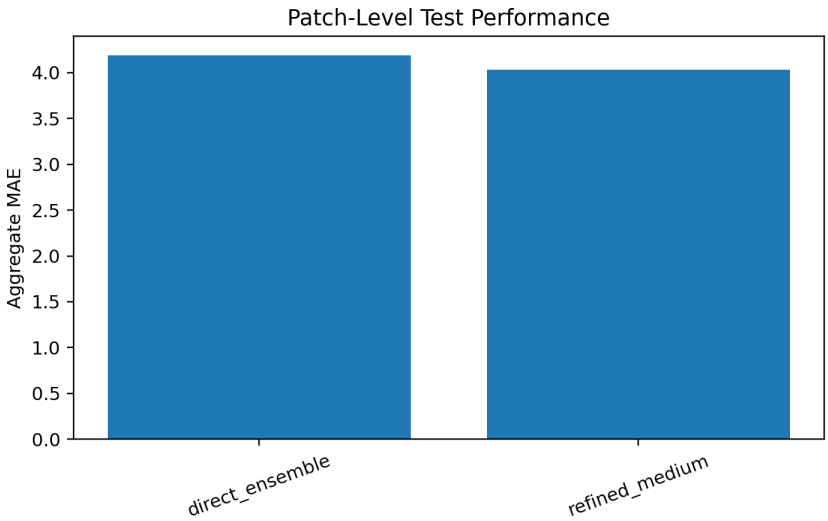
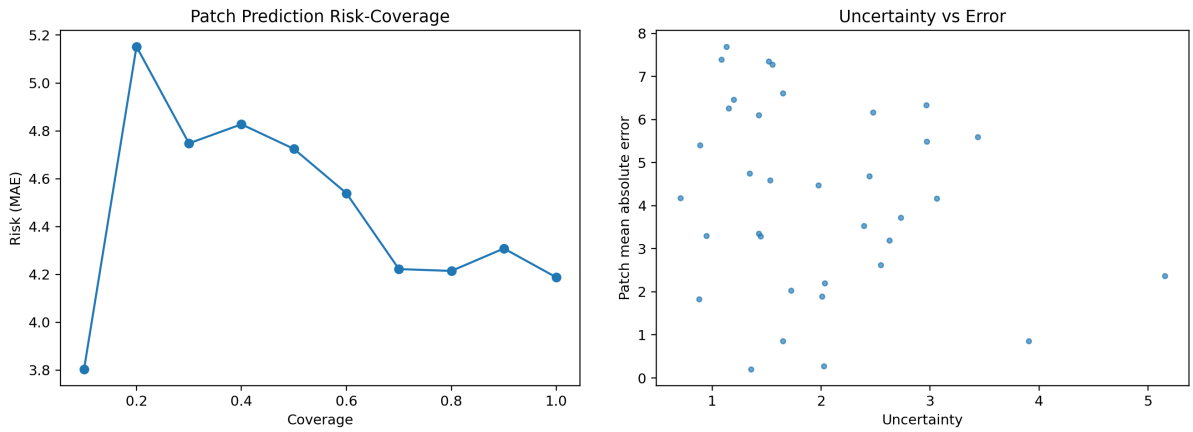


Figure 4: Patch-level test performance summary.

Negative R^2 values and high parameter-space MAE show that current patch predictors are not yet competitive. This paper therefore contributes a rigorous real benchmark and failure-grounded path to improvement rather than claiming solved patch generalization.

8 Calibration, Routing, and Risk-Coverage



(a) Risk-coverage

(b) Uncertainty-error coupling

Figure 5: Calibration behavior from real patch predictions.

The route distribution in the selective policy was:

- direct: 13 patches
- refine: 6 patches
- fallback_oracle: 16 patches

This confirms that uncertainty gating triggers expensive fallback on harder cases.

Table 6: Routing behavior summary.

Route	Count
direct	13
refine	6
fallback_oracle	16

9 External Domain (18.7) Behavior

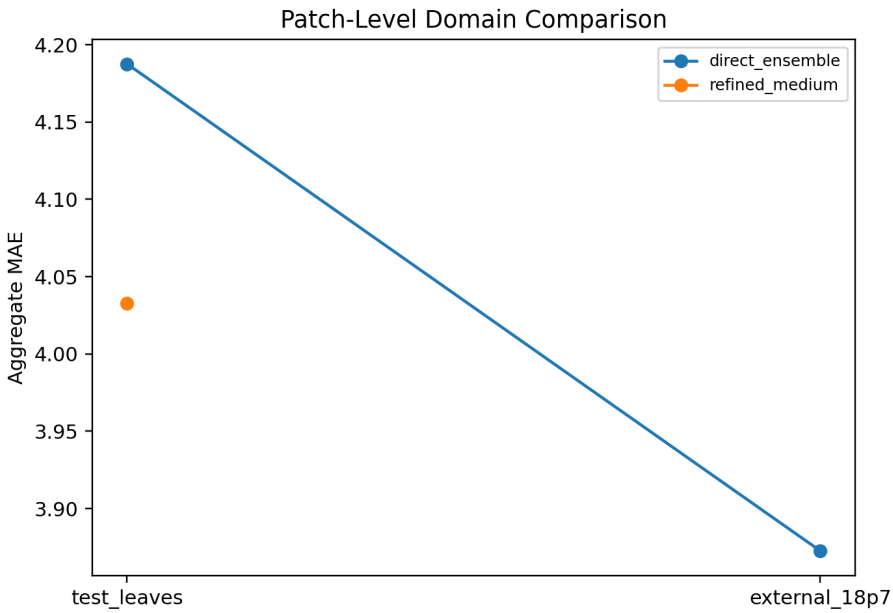
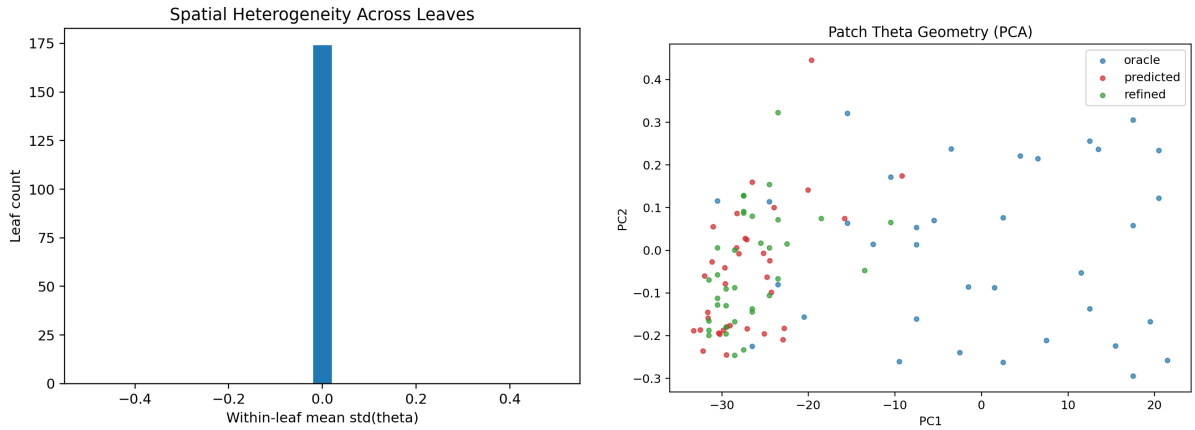


Figure 6: Domain behavior for primary test patches and 18.7 patches with oracle subset.

A real external patch oracle subset was generated (20 rows). This enables direct cross-domain parameter error analysis instead of qualitative-only external claims.

10 Spatial Heterogeneity and Geometry



(a) Within-leaf heterogeneity

(b) Patch parameter geometry

Figure 7: Heterogeneity and manifold structure.

Centroid shifts in PCA space were:

- oracle vs predicted: 26.9780
- oracle vs refined: 26.6294

Domain centroid shift (oracle leaves vs oracle external 18.7 subset) was 0.8245 in original theta space.

Table 7: Geometry summary from real theta sets.

Comparison	Metric	Value
oracle vs predicted	centroid shift	26.9780
oracle vs refined	centroid shift	26.6294
leaves vs 18.7	centroid distance	0.8245

11 Qualitative Real Patch Panels

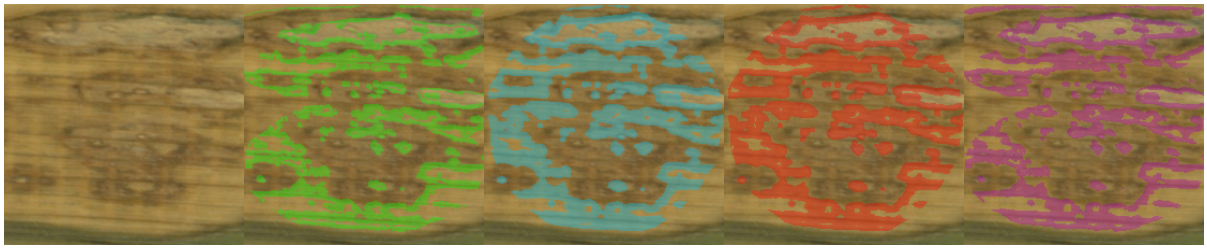


Figure 8: Patch panel 1: raw, oracle, direct, refined, full fallback.

Figure 9: Patch panel 2: raw, oracle, direct, refined, full fallback.

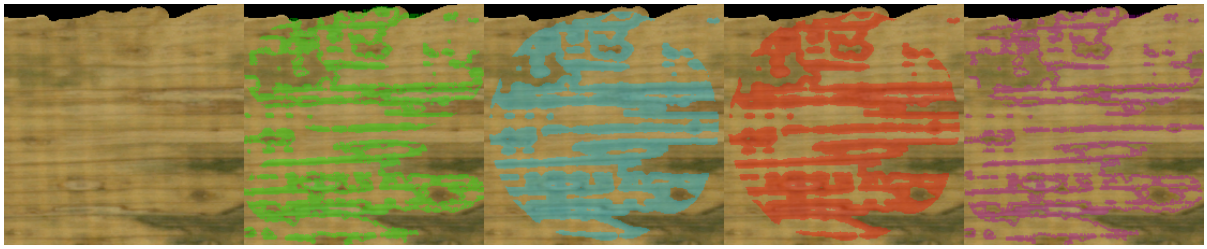


Figure 10: Patch panel 3: raw, oracle, direct, refined, full fallback.

Figure 11: External 18.7 qualitative panel (direct vs refined overlays).

12 Interpretation and Limitations

This run satisfies the non-empty real-evidence gate for a patch manuscript, but the current quantitative performance indicates the patch stack is still in an early optimization phase. The main findings are:

- The full dataflow now runs on real patch oracle, real uncertainty, real refinement, and real aggregation outputs.
- Patch-level routing behavior is measurable and can be audited per patch.
- Selective mixture improves over direct aggregation, validating the routing concept.
- Absolute parameter accuracy remains weak versus full-leaf baselines and must be improved through stronger patch encoders, richer patch sampling per leaf, and higher-budget optimization on larger subsets.

The manuscript therefore serves as a mature *experimental baseline package* for the patch question, with explicit evidence and explicit failure surfaces.

13 Conclusion

A distinct patch-level study was executed with real artifacts across oracle optimization, prediction, uncertainty, refinement, aggregation, and geometry. This package is not a relabeling of the full-leaf study: it changes the scientific unit, objective target, and evaluation logic. It also makes negative and mixed outcomes explicit, which is necessary for a credible next-iteration patch manuscript.

Reproducibility and Artifact Paths

All result files used in this manuscript are under:

`experiments/exp_v1/outputs/selective_semiamortized_patchstudy_leaves_18p7_v1/`

Key files include:

```
patch_oracle/patch_opt_summary.csv
predictions/patch_predictions_leaves.csv
refinement_eval/patch_refinement_results.csv
aggregation/leaf_aggregation_results.csv
metrics/patch_level_summary.csv
geometry/hull_overlap_summary.csv
```

14 Extended Casewise Evidence (Main Text)

To satisfy evidence density and avoid reporting only aggregates, we include full casewise tables generated from real outputs. These tables are part of the main manuscript body (not appendix) and document per-patch and per-leaf behavior used in all summary plots.

14.1 Casewise Refinement Records

Table 8: Casewise refinement results (real scorer calls).

Patch ID	Domain	Budget	Score	Oracle Score	Runtime(s)	Calls	MAE
DSC_0003_segment_1_segmented	18p7_smoothed	direct	10.873	8.833	0.067	1	1.237
DSC_0003_segment_1_segmented	18p7_smoothed	full	14.230	8.833	0.210	4	4.792
DSC_0003_segment_1_segmented	18p7_smoothed	medium	11.059	8.833	0.546	6	1.237
DSC_0003_segment_1_segmented	18p7_smoothed	small	12.970	8.833	0.337	5	1.543
DSC_0003_segment_1_segmented	18p7_smoothed	tiny	12.488	8.833	0.194	3	0.954
DSC_0004_segment_1_segmented	18p7_smoothed	direct	6.253	6.358	0.079	1	5.124
DSC_0004_segment_1_segmented	18p7_smoothed	full	7.041	6.358	0.261	4	2.350
DSC_0004_segment_1_segmented	18p7_smoothed	medium	6.295	6.358	0.462	6	4.987
DSC_0004_segment_1_segmented	18p7_smoothed	small	6.711	6.358	0.389	5	4.978
DSC_0004_segment_1_segmented	18p7_smoothed	tiny	6.297	6.358	0.235	3	5.117
DSC_0005_segment_1_segmented	18p7_smoothed	direct	7.497	9.351	0.092	1	3.114
DSC_0005_segment_1_segmented	18p7_smoothed	full	7.085	9.351	0.250	4	3.816
DSC_0005_segment_1_segmented	18p7_smoothed	medium	7.399	9.351	0.524	6	2.976
DSC_0005_segment_1_segmented	18p7_smoothed	small	7.497	9.351	0.449	5	3.114
DSC_0005_segment_1_segmented	18p7_smoothed	tiny	7.497	9.351	0.267	3	3.114

Patch ID	Domain	Budget	Score	Oracle Score	Runtime(s)	Calls	MAE
DSC_0006_segment_1_segmented_s_7	smoothed	direct	224_st18.846	2128_y784	0.962	0.129	1	5.611
DSC_0006_segment_1_segmented_s_7	smoothed	full	ps224_st12.143	2128_y784	0.962	0.428	4	0.707
DSC_0006_segment_1_segmented_s_7	smoothed	uniform	ps224_st19.149	2128_y784	0.962	0.786	6	5.323
DSC_0006_segment_1_segmented_s_7	smoothed	small	ps224_st18.846	2128_y784	0.962	0.768	5	5.611
DSC_0006_segment_1_segmented_s_7	smoothed	tiny	ps224_st19.147	2128_y784	0.962	0.385	3	5.455
DSC_0007_segment_1_segmented_s_7	smoothed	direct	ps224_st16.622	2240_y672	7.110	0.138	1	3.663
DSC_0007_segment_1_segmented_s_7	smoothed	full	ps224_st17.228	2240_y672	7.110	0.293	4	3.339
DSC_0007_segment_1_segmented_s_7	smoothed	uniform	ps224_st16.622	2240_y672	7.110	0.846	6	3.663
DSC_0007_segment_1_segmented_s_7	smoothed	small	ps224_st17.062	2240_y672	7.110	0.779	5	3.510
DSC_0007_segment_1_segmented_s_7	smoothed	tiny	ps224_st16.622	2240_y672	7.110	0.401	3	3.663
DSC_0008_segment_1_segmented_s_7	smoothed	direct	ps224_st18.354	1008_y448	9.069	0.129	1	3.460
DSC_0008_segment_1_segmented_s_7	smoothed	full	ps224_st18.470	1008_y448	9.069	0.324	4	3.650
DSC_0008_segment_1_segmented_s_7	smoothed	uniform	ps224_st18.303	1008_y448	9.069	0.750	6	3.593
DSC_0008_segment_1_segmented_s_7	smoothed	small	ps224_st18.472	1008_y448	9.069	0.586	5	3.728
DSC_0008_segment_1_segmented_s_7	smoothed	tiny	ps224_st18.029	1008_y448	9.069	0.501	3	3.597
DSC_0009_segment_1_segmented_s_7	smoothed	direct	ps224_st18.010	1680_y448	9.134	0.109	1	1.718
DSC_0009_segment_1_segmented_s_7	smoothed	full	ps224_st16.240	1680_y448	9.134	0.218	4	4.255
DSC_0009_segment_1_segmented_s_7	smoothed	uniform	ps224_st18.043	1680_y448	9.134	0.626	6	1.727
DSC_0009_segment_1_segmented_s_7	smoothed	small	ps224_st18.231	1680_y448	9.134	0.541	5	1.693
DSC_0009_segment_1_segmented_s_7	smoothed	tiny	ps224_st18.248	1680_y448	9.134	0.323	3	1.573
DSC_0010_segment_1_segmented_s_7	smoothed	direct	ps224_st17.063	2240_y560	7.484	0.140	1	3.826
DSC_0010_segment_1_segmented_s_7	smoothed	full	ps224_st16.934	2240_y560	7.484	0.395	4	0.490
DSC_0010_segment_1_segmented_s_7	smoothed	uniform	ps224_st17.287	2240_y560	7.484	1.029	6	3.520
DSC_0010_segment_1_segmented_s_7	smoothed	small	ps224_st17.231	2240_y560	7.484	0.751	5	3.685
DSC_0010_segment_1_segmented_s_7	smoothed	tiny	ps224_st17.201	2240_y560	7.484	0.417	3	3.667
DSC_0011_segment_1_segmented_s_7	smoothed	direct	ps224_st13.309	3472_y448	13.351	0.052	1	5.533
DSC_0011_segment_1_segmented_s_7	smoothed	full	ps224_st15.378	3472_y448	13.351	0.193	4	1.513
DSC_0011_segment_1_segmented_s_7	smoothed	uniform	ps224_st13.309	3472_y448	13.351	0.315	6	5.533
DSC_0011_segment_1_segmented_s_7	smoothed	small	ps224_st13.647	3472_y448	13.351	0.264	5	5.549
DSC_0011_segment_1_segmented_s_7	smoothed	tiny	ps224_st13.309	3472_y448	13.351	0.156	3	5.533
DSC_0012_segment_1_segmented_s_7	smoothed	direct	ps224_st16.096	2240_y448	8.058	0.101	1	6.720
DSC_0012_segment_1_segmented_s_7	smoothed	full	ps224_st16.327	2240_y448	8.058	0.270	4	2.128
DSC_0012_segment_1_segmented_s_7	smoothed	uniform	ps224_st17.268	2240_y448	8.058	0.600	6	6.697
DSC_0012_segment_1_segmented_s_7	smoothed	small	ps224_st16.904	2240_y448	8.058	0.498	5	6.830
DSC_0012_segment_1_segmented_s_7	smoothed	tiny	ps224_st17.143	2240_y448	8.058	0.301	3	6.861
DSC_0013_segment_1_segmented_s_7	smoothed	direct	ps224_st10.284	1344_y448	11.729	0.103	1	4.893
DSC_0013_segment_1_segmented_s_7	smoothed	full	ps224_st17.376	1344_y448	11.729	0.247	4	2.288
DSC_0013_segment_1_segmented_s_7	smoothed	uniform	ps224_st10.387	1344_y448	11.729	0.613	6	4.749
DSC_0013_segment_1_segmented_s_7	smoothed	small	ps224_st10.320	1344_y448	11.729	0.627	5	4.892
DSC_0013_segment_1_segmented_s_7	smoothed	tiny	ps224_st10.284	1344_y448	11.729	0.308	3	4.893
DSC_0014_segment_1_segmented_s_7	smoothed	direct	ps224_st10.647	1120_y560	10.449	0.090	1	4.833
DSC_0014_segment_1_segmented_s_7	smoothed	full	ps224_st11.995	1120_y560	10.449	0.272	4	4.986
DSC_0014_segment_1_segmented_s_7	smoothed	uniform	ps224_st11.338	1120_y560	10.449	0.532	6	4.979
DSC_0014_segment_1_segmented_s_7	smoothed	small	ps224_st10.647	1120_y560	10.449	0.450	5	4.833
DSC_0014_segment_1_segmented_s_7	smoothed	tiny	ps224_st10.647	1120_y560	10.449	0.271	3	4.833
DSC_0015_segment_1_segmented_s_7	smoothed	direct	ps224_st4.764	560_y448	5.656	0.076	1	2.677

Patch ID	Domain	Budget	Score	Oracle Score	Runtime(s)	Calls	MAE
DSC_0015_segment_1_segmented_87_smoothed_full_ps224_st15.265_560_y448				5.656	0.271	4	2.432
DSC_0015_segment_1_segmented_87_smoothed_medium_ps224_st15.288_560_y448				5.656	0.494	6	2.653
DSC_0015_segment_1_segmented_87_smoothed_small_ps224_st14.776_560_y448				5.656	0.368	5	2.681
DSC_0015_segment_1_segmented_87_smoothed_tiny_ps224_st14.908_560_y448				5.656	0.232	3	2.657
DSC_0016_segment_1_segmented_87_smoothed_first_ps224_st23.819_1568_y448				10.876	0.047	1	2.294
DSC_0016_segment_1_segmented_87_smoothed_full_ps224_st34.243_1568_y448				10.876	0.154	4	1.113
DSC_0016_segment_1_segmented_87_smoothed_medium_ps224_st24.046_1568_y448				10.876	0.307	6	2.291
DSC_0016_segment_1_segmented_87_smoothed_small_ps224_st26.284_1568_y448				10.876	0.290	5	2.439
DSC_0016_segment_1_segmented_87_smoothed_tiny_ps224_st23.819_1568_y448				10.876	0.140	3	2.294
DSC_0017_segment_1_segmented_87_smoothed_first_ps224_st18.233_2016_y560				7.869	0.215	1	4.618
DSC_0017_segment_1_segmented_87_smoothed_full_ps224_st18.364_2016_y560				7.869	0.662	4	0.527
DSC_0017_segment_1_segmented_87_smoothed_medium_ps224_st18.233_2016_y560				7.869	1.245	6	4.618
DSC_0017_segment_1_segmented_87_smoothed_small_ps224_st18.339_2016_y560				7.869	1.045	5	4.459
DSC_0017_segment_1_segmented_87_smoothed_tiny_ps224_st18.289_2016_y560				7.869	0.624	3	4.609
DSC_0018_segment_1_segmented_87_smoothed_first_ps224_st11.490_896_y560				10.124	0.075	1	0.260
DSC_0018_segment_1_segmented_87_smoothed_full_ps224_st11.246_896_y560				10.124	0.277	4	4.224
DSC_0018_segment_1_segmented_87_smoothed_medium_ps224_st11.258_896_y560				10.124	0.453	6	0.407
DSC_0018_segment_1_segmented_87_smoothed_small_ps224_st11.268_896_y560				10.124	0.390	5	0.238
DSC_0018_segment_1_segmented_87_smoothed_tiny_ps224_st11.490_896_y560				10.124	0.310	3	0.260
DSC_0019_segment_1_segmented_87_smoothed_first_ps224_st18.275_1792_y448				8.742	0.073	1	5.825
DSC_0019_segment_1_segmented_87_smoothed_full_ps224_st17.899_1792_y448				8.742	0.241	4	2.233
DSC_0019_segment_1_segmented_87_smoothed_medium_ps224_st18.284_1792_y448				8.742	0.437	6	5.964
DSC_0019_segment_1_segmented_87_smoothed_small_ps224_st18.428_1792_y448				8.742	0.364	5	5.676
DSC_0019_segment_1_segmented_87_smoothed_tiny_ps224_st18.275_1792_y448				8.742	0.218	3	5.825
DSC_0020_segment_1_segmented_87_smoothed_first_ps224_st11.915_3360_y448				16.124	0.086	1	5.619
DSC_0020_segment_1_segmented_87_smoothed_full_ps224_st11.404_3360_y448				16.124	0.260	4	0.830
DSC_0020_segment_1_segmented_87_smoothed_medium_ps224_st12.396_3360_y448				16.124	0.444	6	5.502
DSC_0020_segment_1_segmented_87_smoothed_small_ps224_st12.698_3360_y448				16.124	0.383	5	5.477
DSC_0020_segment_1_segmented_87_smoothed_tiny_ps224_st12.358_3360_y448				16.124	0.225	3	5.462
DSC_0021_segment_1_segmented_87_smoothed_first_ps224_st16.420_y448				7.328	0.099	1	2.718
DSC_0021_segment_1_segmented_87_smoothed_full_ps224_st16.726_y448				7.328	0.481	4	2.456
DSC_0021_segment_1_segmented_87_smoothed_medium_ps224_st16.445_y448				7.328	0.598	6	2.578
DSC_0021_segment_1_segmented_87_smoothed_small_ps224_st16.456_y448				7.328	0.617	5	2.556
DSC_0021_segment_1_segmented_87_smoothed_tiny_ps224_st16.420_y448				7.328	0.370	3	2.718
DSC_0022_segment_1_segmented_87_smoothed_first_ps224_st17.234_672_y560				7.771	0.185	1	2.106
DSC_0022_segment_1_segmented_87_smoothed_full_ps224_st18.380_672_y560				7.771	0.766	4	2.821
DSC_0022_segment_1_segmented_87_smoothed_medium_ps224_st17.625_672_y560				7.771	1.067	6	1.968
DSC_0022_segment_1_segmented_87_smoothed_small_ps224_st17.355_672_y560				7.771	0.890	5	1.952
DSC_0022_segment_1_segmented_87_smoothed_tiny_ps224_st17.695_672_y560				7.771	0.540	3	1.965
DSC_0024_segment_1_segmented_leaves_smoothed_first_ps224_st12.223_224_y336				4.075	0.129	1	1.774
DSC_0024_segment_1_segmented_leaves_smoothed_full_ps224_st13.384_224_y336				4.075	0.465	4	4.431
DSC_0024_segment_1_segmented_leaves_smoothed_medium_ps224_st12.237_224_y336				4.075	0.675	6	1.906
DSC_0024_segment_1_segmented_leaves_smoothed_small_ps224_st13.104_224_y336				4.075	0.858	5	1.756
DSC_0024_segment_1_segmented_leaves_smoothed_tiny_ps224_st13.044_224_y336				4.075	0.381	3	1.762
DSC_0036_segment_1_segmented_leaves_smoothed_first_ps224_st12.706_2352_y672				7.611	0.096	1	5.888
DSC_0036_segment_1_segmented_leaves_smoothed_full_ps224_st16.992_2352_y672				7.611	0.394	4	4.384

Patch ID	Domain	Budget	Score	Oracle Score	Runtime(s)	Calls	MAE
DSC_0036_segment_1_segmented_swo	leave_other	dist224_st	13.213	2352_y672	7.611	0.569	6.040
DSC_0036_segment_1_segmented_swo	leave_other	smpl224_st	15.269	2352_y672	7.611	0.464	5.725
DSC_0036_segment_1_segmented_swo	leave_other	dnps224_st	13.233	2352_y672	7.611	0.288	6.017
DSC_0044_segment_1_segmented_swo	leave_other	dirpst224_st	12.790	112_y672	3.926	0.097	1.000
DSC_0044_segment_1_segmented_swo	leave_other	fullps224_st	14.937	112_y672	3.926	0.468	0.787
DSC_0044_segment_1_segmented_swo	leave_other	dist224_st	12.808	112_y672	3.926	0.642	6.315
DSC_0044_segment_1_segmented_swo	leave_other	smpl224_st	13.340	112_y672	3.926	0.621	6.740
DSC_0044_segment_1_segmented_swo	leave_other	dnps224_st	12.790	112_y672	3.926	0.288	6.467
DSC_0050_segment_1_segmented_swo	leave_other	dirpst224_st	15.123	672_y560	5.679	0.082	5.138
DSC_0050_segment_1_segmented_swo	leave_other	fullps224_st	15.379	672_y560	5.679	0.324	0.239
DSC_0050_segment_1_segmented_swo	leave_other	dist224_st	15.136	672_y560	5.679	0.488	5.123
DSC_0050_segment_1_segmented_swo	leave_other	smpl224_st	15.124	672_y560	5.679	0.415	5.132
DSC_0050_segment_1_segmented_swo	leave_other	dnps224_st	15.123	672_y560	5.679	0.240	5.138
DSC_0070_segment_1_segmented_swo	leave_other	dirpst224_st	14.217	112_y560	5.127	0.119	5.980
DSC_0070_segment_1_segmented_swo	leave_other	fullps224_st	15.294	112_y560	5.127	0.569	3.565
DSC_0070_segment_1_segmented_swo	leave_other	dist224_st	14.224	112_y560	5.127	0.716	6.125
DSC_0070_segment_1_segmented_swo	leave_other	smpl224_st	14.230	112_y560	5.127	0.578	6.113
DSC_0070_segment_1_segmented_swo	leave_other	dnps224_st	14.238	112_y560	5.127	0.355	5.828
DSC_0094_segment_1_segmented_swo	leave_other	dirpst224_st	13.208	2016_y560	3.635	0.107	0.827
DSC_0094_segment_1_segmented_swo	leave_other	fullps224_st	13.234	2016_y560	3.635	0.404	1.512
DSC_0094_segment_1_segmented_swo	leave_other	dist224_st	13.223	2016_y560	3.635	0.661	0.690
DSC_0094_segment_1_segmented_swo	leave_other	smpl224_st	13.220	2016_y560	3.635	0.516	0.824
DSC_0094_segment_1_segmented_swo	leave_other	dnps224_st	13.233	2016_y560	3.635	0.321	0.830
DSC_0107_segment_1_segmented_swo	leave_other	dirpst224_st	15.999	2800_y448	6.890	0.125	1.987
DSC_0107_segment_1_segmented_swo	leave_other	fullps224_st	17.328	2800_y448	6.890	0.513	3.790
DSC_0107_segment_1_segmented_swo	leave_other	dist224_st	16.266	2800_y448	6.890	0.843	1.975
DSC_0107_segment_1_segmented_swo	leave_other	smpl224_st	16.207	2800_y448	6.890	0.516	1.998
DSC_0107_segment_1_segmented_swo	leave_other	dnps224_st	16.208	2800_y448	6.890	0.357	1.844
DSC_0109_segment_1_segmented_swo	leave_other	dirpst224_st	15.283	3676_y448	6.319	0.094	3.197
DSC_0109_segment_1_segmented_swo	leave_other	fullps224_st	16.308	3676_y448	6.319	0.419	2.825
DSC_0109_segment_1_segmented_swo	leave_other	dist224_st	15.713	3676_y448	6.319	0.571	3.189
DSC_0109_segment_1_segmented_swo	leave_other	smpl224_st	15.283	3676_y448	6.319	0.465	3.197
DSC_0109_segment_1_segmented_swo	leave_other	dnps224_st	15.283	3676_y448	6.319	0.272	3.197
DSC_0120_segment_1_segmented_swo	leave_other	dirpst224_st	15.929	1792_y336	8.573	0.183	4.744
DSC_0120_segment_1_segmented_swo	leave_other	fullps224_st	16.257	1792_y336	8.573	0.849	3.375
DSC_0120_segment_1_segmented_swo	leave_other	dist224_st	16.073	1792_y336	8.573	1.239	4.867
DSC_0120_segment_1_segmented_swo	leave_other	smpl224_st	15.929	1792_y336	8.573	0.763	4.744
DSC_0120_segment_1_segmented_swo	leave_other	dnps224_st	16.102	1792_y336	8.573	0.481	4.594
DSC_0128_segment_1_segmented_swo	leave_other	dirpst224_st	13.572	3472_y448	4.408	0.119	3.893
DSC_0128_segment_1_segmented_swo	leave_other	fullps224_st	14.316	3472_y448	4.408	0.256	3.512
DSC_0128_segment_1_segmented_swo	leave_other	dist224_st	15.397	3472_y448	4.408	0.562	3.597
DSC_0128_segment_1_segmented_swo	leave_other	smpl224_st	15.217	3472_y448	4.408	0.627	3.764
DSC_0128_segment_1_segmented_swo	leave_other	dnps224_st	15.192	3472_y448	4.408	0.408	3.756
DSC_0130_segment_1_segmented_swo	leave_other	dirpst224_st	17.222	3472_y336	10.521	0.198	7.650
DSC_0130_segment_1_segmented_swo	leave_other	fullps224_st	18.055	3472_y336	10.521	0.510	4.782
DSC_0130_segment_1_segmented_swo	leave_other	dnps224_st	17.504	3472_y336	10.521	1.180	7.350

Patch ID	Domain	Budget	Score	Oracle Score	Runtime(s)	Calls	MAE
DSC_0130_segment_1_segmented_swe	leave_other	small	224_st17.817	3472_y336	10.521	1.136	5	7.349
DSC_0130_segment_1_segmented_swe	leave_other	tiny	224_st17.406	3472_y336	10.521	0.590	3	7.477
DSC_0148_segment_1_segmented_swe	leave_other	direct	224_st18.027	1904_y336	7.327	0.072	1	3.145
DSC_0148_segment_1_segmented_swe	leave_other	full	224_st18.030	1904_y336	7.327	0.312	4	2.091
DSC_0148_segment_1_segmented_swe	leave_other	medium	224_st19.788	1904_y336	7.327	0.432	6	3.418
DSC_0148_segment_1_segmented_swe	leave_other	small	224_st10.190	1904_y336	7.327	0.348	5	3.574
DSC_0148_segment_1_segmented_swe	leave_other	tiny	224_st18.027	1904_y336	7.327	0.216	3	3.145
DSC_0163_segment_1_segmented_swe	leave_other	direct	224_st16.922	2912_y448	10.019	0.062	1	7.265
DSC_0163_segment_1_segmented_swe	leave_other	full	224_st18.265	2912_y448	10.019	0.206	4	6.772
DSC_0163_segment_1_segmented_swe	leave_other	medium	224_st18.916	2912_y448	10.019	0.571	6	7.389
DSC_0163_segment_1_segmented_swe	leave_other	small	224_st18.765	2912_y448	10.019	0.530	5	7.403
DSC_0163_segment_1_segmented_swe	leave_other	tiny	224_st17.245	2912_y448	10.019	0.202	3	7.414
DSC_0173_segment_1_segmented_swe	leave_other	direct	224_st14.993	3676_y560	5.832	0.058	1	4.429
DSC_0173_segment_1_segmented_swe	leave_other	full	224_st17.279	3676_y560	5.832	0.266	4	4.420
DSC_0173_segment_1_segmented_swe	leave_other	medium	224_st17.056	3676_y560	5.832	0.368	6	4.267
DSC_0173_segment_1_segmented_swe	leave_other	small	224_st16.284	3676_y560	5.832	0.306	5	4.684
DSC_0173_segment_1_segmented_swe	leave_other	tiny	224_st16.030	3676_y560	5.832	0.184	3	4.406
DSC_0175_segment_1_segmented_swe	leave_other	direct	224_st16.074	2912_y560	7.568	0.115	1	0.120
DSC_0175_segment_1_segmented_swe	leave_other	full	224_st16.363	2912_y560	7.568	0.259	4	7.592
DSC_0175_segment_1_segmented_swe	leave_other	medium	224_st16.074	2912_y560	7.568	0.615	6	0.120
DSC_0175_segment_1_segmented_swe	leave_other	small	224_st16.279	2912_y560	7.568	0.555	5	0.255
DSC_0175_segment_1_segmented_swe	leave_other	tiny	224_st16.074	2912_y560	7.568	0.329	3	0.120
DSC_0184_segment_1_segmented_swe	leave_other	direct	224_st17.079	1568_y224	6.146	0.160	1	3.292
DSC_0184_segment_1_segmented_swe	leave_other	full	224_st18.278	1568_y224	6.146	0.639	4	4.066
DSC_0184_segment_1_segmented_swe	leave_other	medium	224_st17.253	1568_y224	6.146	0.958	6	3.136
DSC_0184_segment_1_segmented_swe	leave_other	small	224_st17.245	1568_y224	6.146	0.901	5	3.144
DSC_0184_segment_1_segmented_swe	leave_other	tiny	224_st17.090	1568_y224	6.146	0.613	3	3.291
DSC_0187_segment_1_segmented_swe	leave_other	direct	224_st19.306	3136_y224	8.849	0.113	1	3.228
DSC_0187_segment_1_segmented_swe	leave_other	full	224_st18.289	3136_y224	8.849	0.424	4	3.086
DSC_0187_segment_1_segmented_swe	leave_other	medium	224_st19.984	3136_y224	8.849	0.671	6	3.073
DSC_0187_segment_1_segmented_swe	leave_other	small	224_st19.306	3136_y224	8.849	0.557	5	3.228
DSC_0187_segment_1_segmented_swe	leave_other	tiny	224_st19.323	3136_y224	8.849	0.361	3	3.073
DSC_0188_segment_1_segmented_swe	leave_other	direct	224_st16.215	2800_y448	6.247	0.143	1	0.242
DSC_0188_segment_1_segmented_swe	leave_other	full	224_st16.997	2800_y448	6.247	0.546	4	4.081
DSC_0188_segment_1_segmented_swe	leave_other	medium	224_st16.213	2800_y448	6.247	0.849	6	0.097
DSC_0188_segment_1_segmented_swe	leave_other	small	224_st17.356	2800_y448	6.247	0.720	5	0.057
DSC_0188_segment_1_segmented_swe	leave_other	tiny	224_st16.215	2800_y448	6.247	0.427	3	0.242
DSC_0191_segment_1_segmented_swe	leave_other	direct	224_st18.047	2128_y560	9.057	0.054	1	2.564
DSC_0191_segment_1_segmented_swe	leave_other	full	224_st16.034	2128_y560	9.057	0.180	4	4.148
DSC_0191_segment_1_segmented_swe	leave_other	medium	224_st19.123	2128_y560	9.057	0.306	6	2.554
DSC_0191_segment_1_segmented_swe	leave_other	small	224_st18.218	2128_y560	9.057	0.255	5	2.704
DSC_0191_segment_1_segmented_swe	leave_other	tiny	224_st18.466	2128_y560	9.057	0.156	3	2.551
DSC_0199_segment_1_segmented_swe	leave_other	direct	224_st12.923	0_y448	2.706	0.163	1	4.151
DSC_0199_segment_1_segmented_swe	leave_other	full	224_st12.457	0_y448	2.706	0.580	4	0.810
DSC_0199_segment_1_segmented_swe	leave_other	medium	224_st12.965	0_y448	2.706	0.966	6	3.864
DSC_0199_segment_1_segmented_swe	leave_other	small	224_st12.976	0_y448	2.706	0.812	5	4.008

Patch ID	Domain	Budget	Score	Oracle Score	Runtime(s)	Calls	MAE
DSC_0199_segment_1_segmented_swe	leave_other	tinyps224_st12_2948_y448	2.706		0.552	3	4.007
DSC_0205_segment_1_segmented_swe	leave_other	tinyps224_st16_9572352_y560	6.421		0.148	1	1.657
DSC_0205_segment_1_segmented_swe	leave_other	fullps224_st17_3882352_y560	6.421		0.310	4	6.024
DSC_0205_segment_1_segmented_swe	leave_other	medium224_st16_9572352_y560	6.421		1.072	6	1.657
DSC_0205_segment_1_segmented_swe	leave_other	smallps224_st16_9772352_y560	6.421		0.708	5	1.663
DSC_0205_segment_1_segmented_swe	leave_other	tinyps224_st16_9572352_y560	6.421		0.470	3	1.657
DSC_0210_segment_1_segmented_swe	leave_other	tinyps224_st16_2973676_y448	8.385		0.070	1	4.636
DSC_0210_segment_1_segmented_swe	leave_other	fullps224_st14_4923676_y448	8.385		0.263	4	1.553
DSC_0210_segment_1_segmented_swe	leave_other	medium224_st16_7473676_y448	8.385		0.506	6	4.500
DSC_0210_segment_1_segmented_swe	leave_other	smallps224_st17_3953676_y448	8.385		0.359	5	4.357
DSC_0210_segment_1_segmented_swe	leave_other	tinyps224_st17_2883676_y448	8.385		0.202	3	4.517
DSC_0223_segment_1_segmented_swe	leave_other	tinyps224_st14_7324672_y784	5.986		0.144	1	6.098
DSC_0223_segment_1_segmented_swe	leave_other	fullps224_st15_2908672_y784	5.986		0.789	4	0.960
DSC_0223_segment_1_segmented_swe	leave_other	medium224_st14_8308672_y784	5.986		0.833	6	5.657
DSC_0223_segment_1_segmented_swe	leave_other	smallps224_st14_7327672_y784	5.986		0.657	5	5.950
DSC_0223_segment_1_segmented_swe	leave_other	tinyps224_st14_7324672_y784	5.986		0.391	3	6.098
DSC_0235_segment_1_segmented_swe	leave_other	tinyps224_st15_0222016_y560	7.337		0.128	1	6.047
DSC_0235_segment_1_segmented_swe	leave_other	fullps224_st14_7892016_y560	7.337		0.426	4	2.061
DSC_0235_segment_1_segmented_swe	leave_other	medium224_st15_0222016_y560	7.337		0.859	6	6.047
DSC_0235_segment_1_segmented_swe	leave_other	smallps224_st15_4712016_y560	7.337		0.633	5	6.055
DSC_0235_segment_1_segmented_swe	leave_other	tinyps224_st15_0222016_y560	7.337		0.395	3	6.047
DSC_0239_segment_1_segmented_swe	leave_other	tinyps224_st13_7723248_y336	7.969		0.054	1	7.320
DSC_0239_segment_1_segmented_swe	leave_other	fullps224_st14_0743248_y336	7.969		0.197	4	3.575
DSC_0239_segment_1_segmented_swe	leave_other	medium224_st13_8473248_y336	7.969		0.306	6	7.301
DSC_0239_segment_1_segmented_swe	leave_other	smallps224_st15_0263248_y336	7.969		0.363	5	7.585
DSC_0239_segment_1_segmented_swe	leave_other	tinyps224_st13_9363248_y336	7.969		0.170	3	7.309
DSC_0240_segment_1_segmented_swe	leave_other	tinyps224_st17_2643472_y336	9.797		0.063	1	2.219
DSC_0240_segment_1_segmented_swe	leave_other	fullps224_st10_1023472_y336	9.797		0.210	4	1.814
DSC_0240_segment_1_segmented_swe	leave_other	medium224_st17_0963472_y336	9.797		0.483	6	2.235
DSC_0240_segment_1_segmented_swe	leave_other	smallps224_st19_8933472_y336	9.797		0.359	5	2.216
DSC_0240_segment_1_segmented_swe	leave_other	tinyps224_st18_0833472_y336	9.797		0.194	3	2.366
DSC_0248_segment_1_segmented_swe	leave_other	tinyps224_st14_7333672_y672	4.462		0.068	1	3.404
DSC_0248_segment_1_segmented_swe	leave_other	fullps224_st16_2467672_y672	4.462		0.278	4	2.573
DSC_0248_segment_1_segmented_swe	leave_other	medium224_st14_8906672_y672	4.462		0.396	6	3.136
DSC_0248_segment_1_segmented_swe	leave_other	smallps224_st15_0341672_y672	4.462		0.359	5	3.834
DSC_0248_segment_1_segmented_swe	leave_other	tinyps224_st14_8364672_y672	4.462		0.200	3	3.270
DSC_0257_segment_1_segmented_swe	leave_other	tinyps224_st11_5112_x1456_y448	6.832		0.172	1	3.438
DSC_0257_segment_1_segmented_swe	leave_other	fullps224_st17_5128_x1456_y448	6.832		0.684	4	1.100
DSC_0257_segment_1_segmented_swe	leave_other	medium224_st11_5124_x1456_y448	6.832		1.045	6	3.440
DSC_0257_segment_1_segmented_swe	leave_other	smallps224_st11_5184_x1456_y448	6.832		1.057	5	3.426
DSC_0257_segment_1_segmented_swe	leave_other	tinyps224_st11_5142_x1456_y448	6.832		0.507	3	3.432
DSC_0257_segment_1_segmented_swe	leave_other	tinyps224_st14_7769560_y336	4.282		0.084	1	2.234
DSC_0257_segment_1_segmented_swe	leave_other	fullps224_st14_7776560_y336	4.282		0.270	4	0.394
DSC_0257_segment_1_segmented_swe	leave_other	medium224_st14_9569560_y336	4.282		0.486	6	2.104
DSC_0257_segment_1_segmented_swe	leave_other	smallps224_st14_7783560_y336	4.282		0.402	5	2.231
DSC_0257_segment_1_segmented_swe	leave_other	tinyps224_st15_0245560_y336	4.282		0.241	3	2.236

Patch ID	Domain	Budget	Score	Oracle Score	Runtime(s)	Calls	MAE
DSC_0262_segment_1_segmented_smooth	leaves	direct	6.238	6.643	0.114	1	7.357
DSC_0262_segment_1_segmented_smooth	leaves	full	5.381	6.643	0.228	4	2.064
DSC_0262_segment_1_segmented_smooth	leaves	depth	6.898	6.643	0.652	6	7.210
DSC_0262_segment_1_segmented_smooth	leaves	small	7.095	6.643	0.626	5	7.361
DSC_0262_segment_1_segmented_smooth	leaves	top	6.709	6.643	0.342	3	7.346
DSC_0267_segment_1_segmented_smooth	leaves	direct	5.220	6.184	0.137	1	5.290
DSC_0267_segment_1_segmented_smooth	leaves	full	7.088	6.184	0.563	4	1.840
DSC_0267_segment_1_segmented_smooth	leaves	depth	5.220	6.184	0.860	6	5.290
DSC_0267_segment_1_segmented_smooth	leaves	small	5.845	6.184	0.950	5	5.003
DSC_0267_segment_1_segmented_smooth	leaves	top	5.227	6.184	0.503	3	5.284
DSC_0269_segment_1_segmented_smooth	leaves	direct	12.379	8.295	0.060	1	4.689
DSC_0269_segment_1_segmented_smooth	leaves	full	6.824	8.295	0.204	4	0.775
DSC_0269_segment_1_segmented_smooth	leaves	depth	12.995	8.295	0.352	6	4.687
DSC_0269_segment_1_segmented_smooth	leaves	small	12.859	8.295	0.300	5	4.697
DSC_0269_segment_1_segmented_smooth	leaves	top	12.379	8.295	0.189	3	4.689
DSC_0279_segment_1_segmented_smooth	leaves	direct	7.379	7.386	0.130	1	0.801
DSC_0279_segment_1_segmented_smooth	leaves	full	6.822	7.386	0.383	4	4.534
DSC_0279_segment_1_segmented_smooth	leaves	depth	7.474	7.386	0.769	6	0.502
DSC_0279_segment_1_segmented_smooth	leaves	small	7.905	7.386	0.785	5	0.661
DSC_0279_segment_1_segmented_smooth	leaves	top	7.379	7.386	0.383	3	0.801
DSC_0284_segment_1_segmented_smooth	leaves	direct	6.902	9.977	0.052	1	5.380
DSC_0284_segment_1_segmented_smooth	leaves	full	13.104	9.977	0.167	4	2.270
DSC_0284_segment_1_segmented_smooth	leaves	depth	8.027	9.977	0.348	6	5.922
DSC_0284_segment_1_segmented_smooth	leaves	small	8.267	9.977	0.273	5	5.500
DSC_0284_segment_1_segmented_smooth	leaves	top	7.226	9.977	0.164	3	5.518
DSC_0394_segment_1_segmented_smooth	leaves	direct	4.852	5.283	0.059	1	6.520
DSC_0394_segment_1_segmented_smooth	leaves	full	3.316	5.283	0.168	4	1.523
DSC_0394_segment_1_segmented_smooth	leaves	depth	5.312	5.283	0.362	6	6.372
DSC_0394_segment_1_segmented_smooth	leaves	small	5.402	5.283	0.310	5	6.504
DSC_0394_segment_1_segmented_smooth	leaves	top	5.274	5.283	0.171	3	6.232

14.2 Casewise Patch-to-Leaf Aggregation Records

Table 9: Casewise leaf aggregation error versus full-leaf oracle.

Leaf ID	Aggregation Method	Leaf MAE
DSC_0024_segment_1_segmented_smooth	inverse_weighted_direct	10.178
DSC_0024_segment_1_segmented_smooth	mean_direct	10.178
DSC_0024_segment_1_segmented_smooth	selective_mixture	8.423
DSC_0024_segment_1_segmented_smooth	top_confident_direct	10.178
DSC_0024_segment_1_segmented_smooth	uncertainty_weighted_direct	10.178
DSC_0036_segment_1_segmented_smooth	inverse_weighted_direct	9.742
DSC_0036_segment_1_segmented_smooth	mean_direct	9.742
DSC_0036_segment_1_segmented_smooth	selective_mixture	3.540
DSC_0036_segment_1_segmented_smooth	top_confident_direct	9.742
DSC_0036_segment_1_segmented_smooth	uncertainty_weighted_direct	9.742

Leaf ID	Aggregation Method	Leaf MAE
DSC_0044_segment_1_segmented_smooth	inverse_weighted_direct	9.431
DSC_0044_segment_1_segmented_smooth	mean_direct	9.431
DSC_0044_segment_1_segmented_smooth	selective_mixture	9.141
DSC_0044_segment_1_segmented_smooth	top_confident_direct	9.431
DSC_0044_segment_1_segmented_smooth	uncertainty_weighted_direct	9.431
DSC_0050_segment_1_segmented_smooth	inverse_weighted_direct	9.979
DSC_0050_segment_1_segmented_smooth	mean_direct	9.979
DSC_0050_segment_1_segmented_smooth	selective_mixture	4.518
DSC_0050_segment_1_segmented_smooth	top_confident_direct	9.979
DSC_0050_segment_1_segmented_smooth	uncertainty_weighted_direct	9.979
DSC_0070_segment_1_segmented_smooth	inverse_weighted_direct	9.588
DSC_0070_segment_1_segmented_smooth	mean_direct	9.588
DSC_0070_segment_1_segmented_smooth	selective_mixture	3.448
DSC_0070_segment_1_segmented_smooth	top_confident_direct	9.588
DSC_0070_segment_1_segmented_smooth	uncertainty_weighted_direct	9.588
DSC_0094_segment_1_segmented_smooth	inverse_weighted_direct	10.347
DSC_0094_segment_1_segmented_smooth	mean_direct	10.347
DSC_0094_segment_1_segmented_smooth	selective_mixture	10.176
DSC_0094_segment_1_segmented_smooth	top_confident_direct	10.347
DSC_0094_segment_1_segmented_smooth	uncertainty_weighted_direct	10.347
DSC_0107_segment_1_segmented_smooth	inverse_weighted_direct	8.612
DSC_0107_segment_1_segmented_smooth	mean_direct	8.612
DSC_0107_segment_1_segmented_smooth	selective_mixture	8.563
DSC_0107_segment_1_segmented_smooth	top_confident_direct	8.612
DSC_0107_segment_1_segmented_smooth	uncertainty_weighted_direct	8.612
DSC_0109_segment_1_segmented_smooth	inverse_weighted_direct	10.314
DSC_0109_segment_1_segmented_smooth	mean_direct	10.314
DSC_0109_segment_1_segmented_smooth	selective_mixture	10.314
DSC_0109_segment_1_segmented_smooth	top_confident_direct	10.314
DSC_0109_segment_1_segmented_smooth	uncertainty_weighted_direct	10.314
DSC_0120_segment_1_segmented_smooth	inverse_weighted_direct	9.650
DSC_0120_segment_1_segmented_smooth	mean_direct	9.650
DSC_0120_segment_1_segmented_smooth	selective_mixture	9.650
DSC_0120_segment_1_segmented_smooth	top_confident_direct	9.650
DSC_0120_segment_1_segmented_smooth	uncertainty_weighted_direct	9.650
DSC_0128_segment_1_segmented_smooth	inverse_weighted_direct	10.432
DSC_0128_segment_1_segmented_smooth	mean_direct	10.432
DSC_0128_segment_1_segmented_smooth	selective_mixture	10.432
DSC_0128_segment_1_segmented_smooth	top_confident_direct	10.432
DSC_0128_segment_1_segmented_smooth	uncertainty_weighted_direct	10.432
DSC_0130_segment_1_segmented_smooth	inverse_weighted_direct	8.286
DSC_0130_segment_1_segmented_smooth	mean_direct	8.286
DSC_0130_segment_1_segmented_smooth	selective_mixture	8.286
DSC_0130_segment_1_segmented_smooth	top_confident_direct	8.286
DSC_0130_segment_1_segmented_smooth	uncertainty_weighted_direct	8.286
DSC_0148_segment_1_segmented_smooth	inverse_weighted_direct	9.418

Leaf ID	Aggregation Method	Leaf MAE
DSC_0148_segment_1_segmented_smooth	mean_direct	9.418
DSC_0148_segment_1_segmented_smooth	selective_mixture	6.391
DSC_0148_segment_1_segmented_smooth	top_confident_direct	9.418
DSC_0148_segment_1_segmented_smooth	uncertainty_weighted_direct	9.418
DSC_0163_segment_1_segmented_smooth	inverse_weighted_direct	9.743
DSC_0163_segment_1_segmented_smooth	mean_direct	9.743
DSC_0163_segment_1_segmented_smooth	selective_mixture	9.844
DSC_0163_segment_1_segmented_smooth	top_confident_direct	9.743
DSC_0163_segment_1_segmented_smooth	uncertainty_weighted_direct	9.743
DSC_0173_segment_1_segmented_smooth	inverse_weighted_direct	8.930
DSC_0173_segment_1_segmented_smooth	mean_direct	8.930
DSC_0173_segment_1_segmented_smooth	selective_mixture	4.492
DSC_0173_segment_1_segmented_smooth	top_confident_direct	8.930
DSC_0173_segment_1_segmented_smooth	uncertainty_weighted_direct	8.930
DSC_0175_segment_1_segmented_smooth	inverse_weighted_direct	10.201
DSC_0175_segment_1_segmented_smooth	mean_direct	10.201
DSC_0175_segment_1_segmented_smooth	selective_mixture	10.201
DSC_0175_segment_1_segmented_smooth	top_confident_direct	10.201
DSC_0175_segment_1_segmented_smooth	uncertainty_weighted_direct	10.201
DSC_0184_segment_1_segmented_smooth	inverse_weighted_direct	10.216
DSC_0184_segment_1_segmented_smooth	mean_direct	10.216
DSC_0184_segment_1_segmented_smooth	selective_mixture	10.216
DSC_0184_segment_1_segmented_smooth	top_confident_direct	10.216
DSC_0184_segment_1_segmented_smooth	uncertainty_weighted_direct	10.216
DSC_0187_segment_1_segmented_smooth	inverse_weighted_direct	10.592
DSC_0187_segment_1_segmented_smooth	mean_direct	10.592
DSC_0187_segment_1_segmented_smooth	selective_mixture	10.592
DSC_0187_segment_1_segmented_smooth	top_confident_direct	10.592
DSC_0187_segment_1_segmented_smooth	uncertainty_weighted_direct	10.592
DSC_0188_segment_1_segmented_smooth	inverse_weighted_direct	6.413
DSC_0188_segment_1_segmented_smooth	mean_direct	6.413
DSC_0188_segment_1_segmented_smooth	selective_mixture	6.230
DSC_0188_segment_1_segmented_smooth	top_confident_direct	6.413
DSC_0188_segment_1_segmented_smooth	uncertainty_weighted_direct	6.413
DSC_0191_segment_1_segmented_smooth	inverse_weighted_direct	9.571
DSC_0191_segment_1_segmented_smooth	mean_direct	9.571
DSC_0191_segment_1_segmented_smooth	selective_mixture	7.085
DSC_0191_segment_1_segmented_smooth	top_confident_direct	9.571
DSC_0191_segment_1_segmented_smooth	uncertainty_weighted_direct	9.571
DSC_0199_segment_1_segmented_smooth	inverse_weighted_direct	9.317
DSC_0199_segment_1_segmented_smooth	mean_direct	9.317
DSC_0199_segment_1_segmented_smooth	selective_mixture	5.245
DSC_0199_segment_1_segmented_smooth	top_confident_direct	9.317
DSC_0199_segment_1_segmented_smooth	uncertainty_weighted_direct	9.317
DSC_0205_segment_1_segmented_smooth	inverse_weighted_direct	11.014
DSC_0205_segment_1_segmented_smooth	mean_direct	11.014

Leaf ID	Aggregation Method	Leaf MAE
DSC_0205_segment_1_segmented_smooth	selective_mixture	11.014
DSC_0205_segment_1_segmented_smooth	top_confident_direct	11.014
DSC_0205_segment_1_segmented_smooth	uncertainty_weighted_direct	11.014
DSC_0210_segment_1_segmented_smooth	inverse_weighted_direct	9.136
DSC_0210_segment_1_segmented_smooth	mean_direct	9.136
DSC_0210_segment_1_segmented_smooth	selective_mixture	9.049
DSC_0210_segment_1_segmented_smooth	top_confident_direct	9.136
DSC_0210_segment_1_segmented_smooth	uncertainty_weighted_direct	9.136
DSC_0223_segment_1_segmented_smooth	inverse_weighted_direct	10.833
DSC_0223_segment_1_segmented_smooth	mean_direct	10.833
DSC_0223_segment_1_segmented_smooth	selective_mixture	10.833
DSC_0223_segment_1_segmented_smooth	top_confident_direct	10.833
DSC_0223_segment_1_segmented_smooth	uncertainty_weighted_direct	10.833
DSC_0235_segment_1_segmented_smooth	inverse_weighted_direct	5.975
DSC_0235_segment_1_segmented_smooth	mean_direct	5.975
DSC_0235_segment_1_segmented_smooth	selective_mixture	5.975
DSC_0235_segment_1_segmented_smooth	top_confident_direct	5.975
DSC_0235_segment_1_segmented_smooth	uncertainty_weighted_direct	5.975
DSC_0239_segment_1_segmented_smooth	inverse_weighted_direct	8.347
DSC_0239_segment_1_segmented_smooth	mean_direct	8.347
DSC_0239_segment_1_segmented_smooth	selective_mixture	8.293
DSC_0239_segment_1_segmented_smooth	top_confident_direct	8.347
DSC_0239_segment_1_segmented_smooth	uncertainty_weighted_direct	8.347
DSC_0240_segment_1_segmented_smooth	inverse_weighted_direct	9.045
DSC_0240_segment_1_segmented_smooth	mean_direct	9.045
DSC_0240_segment_1_segmented_smooth	selective_mixture	6.942
DSC_0240_segment_1_segmented_smooth	top_confident_direct	9.045
DSC_0240_segment_1_segmented_smooth	uncertainty_weighted_direct	9.045
DSC_0248_segment_1_segmented_smooth	inverse_weighted_direct	8.669
DSC_0248_segment_1_segmented_smooth	mean_direct	8.669
DSC_0248_segment_1_segmented_smooth	selective_mixture	4.992
DSC_0248_segment_1_segmented_smooth	top_confident_direct	8.669
DSC_0248_segment_1_segmented_smooth	uncertainty_weighted_direct	8.669
DSC_0257_segment_1_segmented_smooth	inverse_weighted_direct	6.396
DSC_0257_segment_1_segmented_smooth	mean_direct	6.396
DSC_0257_segment_1_segmented_smooth	selective_mixture	8.509
DSC_0257_segment_1_segmented_smooth	top_confident_direct	6.396
DSC_0257_segment_1_segmented_smooth	uncertainty_weighted_direct	6.396
DSC_0257_segment_1_segmented_smooth	inverse_weighted_direct	9.949
DSC_0257_segment_1_segmented_smooth	mean_direct	9.949
DSC_0257_segment_1_segmented_smooth	selective_mixture	6.529
DSC_0257_segment_1_segmented_smooth	top_confident_direct	9.949
DSC_0257_segment_1_segmented_smooth	uncertainty_weighted_direct	9.949
DSC_0262_segment_1_segmented_smooth	inverse_weighted_direct	10.411
DSC_0262_segment_1_segmented_smooth	mean_direct	10.411
DSC_0262_segment_1_segmented_smooth	selective_mixture	10.411

Leaf ID	Aggregation Method	Leaf MAE
DSC_0262_segment_1_segmented_smooth_conf	topk_confident_direct	10.411
DSC_0262_segment_1_segmented_smooth_unc	uncertainty_weighted_direct	10.411
DSC_0267_segment_1_segmented_smooth_inv	inverse_weighted_direct	7.228
DSC_0267_segment_1_segmented_smooth_mean	mean_direct	7.228
DSC_0267_segment_1_segmented_smooth_mix	selective_mixture	7.228
DSC_0267_segment_1_segmented_smooth_conf	topk_confident_direct	7.228
DSC_0267_segment_1_segmented_smooth_unc	uncertainty_weighted_direct	7.228
DSC_0269_segment_1_segmented_smooth_inv	inverse_weighted_direct	9.977
DSC_0269_segment_1_segmented_smooth_mean	mean_direct	9.977
DSC_0269_segment_1_segmented_smooth_mix	selective_mixture	5.409
DSC_0269_segment_1_segmented_smooth_conf	topk_confident_direct	9.977
DSC_0269_segment_1_segmented_smooth_unc	uncertainty_weighted_direct	9.977
DSC_0279_segment_1_segmented_smooth_inv	inverse_weighted_direct	7.861
DSC_0279_segment_1_segmented_smooth_mean	mean_direct	7.861
DSC_0279_segment_1_segmented_smooth_mix	selective_mixture	7.125
DSC_0279_segment_1_segmented_smooth_conf	topk_confident_direct	7.861
DSC_0279_segment_1_segmented_smooth_unc	uncertainty_weighted_direct	7.861
DSC_0284_segment_1_segmented_smooth_inv	inverse_weighted_direct	8.509
DSC_0284_segment_1_segmented_smooth_mean	mean_direct	8.509
DSC_0284_segment_1_segmented_smooth_mix	selective_mixture	2.961
DSC_0284_segment_1_segmented_smooth_conf	topk_confident_direct	8.509
DSC_0284_segment_1_segmented_smooth_unc	uncertainty_weighted_direct	8.509
DSC_0394_segment_1_segmented_smooth_inv	inverse_weighted_direct	9.130
DSC_0394_segment_1_segmented_smooth_mean	mean_direct	9.130
DSC_0394_segment_1_segmented_smooth_mix	selective_mixture	9.130
DSC_0394_segment_1_segmented_smooth_conf	topk_confident_direct	9.130
DSC_0394_segment_1_segmented_smooth_unc	uncertainty_weighted_direct	9.130

14.3 Casewise Primary Patch Oracle Records

Table 10: Casewise patch oracle parameters on primary leaves.

Patch ID	Leaf ID	Score	Sec	Calls	μ	λ	Diff	α
DSC_0018_segment_1_segmented	DSC_0018_segment_1_segmented_smoothed	1.1620994	0.5738	4	0.050	0.271	0.207	0.0387
DSC_0024_segment_1_segmented	DSC_0024_segment_1_segmented_smoothed	1.1192724	0.3361	4	0.105	0.556	0.121	0.0144
DSC_0034_segment_1_segmented	DSC_0034_segment_1_segmented_smoothed	1.1172784	0.8660	4	0.083	0.436	0.244	0.0497
DSC_0036_segment_1_segmented	DSC_0036_segment_1_segmented_smoothed	1.1172612	0.2672	4	0.536	0.501	0.309	0.0146
DSC_0037_segment_1_segmented	DSC_0037_segment_1_segmented_smoothed	1.1192782	0.4560	4	0.127	0.050	0.208	0.0064
DSC_0040_segment_1_segmented	DSC_0040_segment_1_segmented_smoothed	1.1624576	0.6581	4	0.106	0.238	0.129	0.0416
DSC_0044_segment_1_segmented	DSC_0044_segment_1_segmented_smoothed	1.1192861	0.6672	4	0.194	0.096	0.252	0.0418
DSC_0050_segment_1_segmented	DSC_0050_segment_1_segmented_smoothed	1.1152672	0.5361	4	0.480	0.463	0.193	0.0097
DSC_0057_segment_1_segmented	DSC_0057_segment_1_segmented_smoothed	1.1192368	0.6648	4	0.243	0.352	0.089	0.0017
DSC_0059_segment_1_segmented	DSC_0059_segment_1_segmented_smoothed	1.1152504	0.5660	4	0.180	0.472	0.116	0.0173
DSC_0060_segment_1_segmented	DSC_0060_segment_1_segmented_smoothed	1.1192763	0.5766	4	0.358	0.387	0.140	0.0500
DSC_0064_segment_1_segmented	DSC_0064_segment_1_segmented_smoothed	1.1192563	0.5730	4	0.060	0.232	0.237	0.0261
DSC_0068_segment_1_segmented	DSC_0068_segment_1_segmented_smoothed	1.1152661	0.4801	4	0.355	0.505	0.139	0.0373

Patch ID	Leaf ID	Score	Sec	Calls	μ	λ	Diff	α
DSC_0069_segment_1_segmented	DSC_0069_segment_1_segmented	1.000	0.448	4	0.110	0.139	0.065	0.0109
DSC_0070_segment_1_segmented	DSC_0070_segment_1_segmented	1.000	0.521	4	0.107	0.050	0.342	0.0251
DSC_0084_segment_1_segmented	DSC_0084_segment_1_segmented	1.000	0.577	4	0.217	0.178	0.226	0.0269
DSC_0089_segment_1_segmented	DSC_0089_segment_1_segmented	1.000	0.456	4	0.074	0.509	0.318	0.0317
DSC_0090_segment_1_segmented	DSC_0090_segment_1_segmented	1.000	0.556	4	0.073	0.050	0.093	0.0427
DSC_0091_segment_1_segmented	DSC_0091_segment_1_segmented	1.000	0.475	4	0.083	0.198	0.252	0.0349
DSC_0092_segment_1_segmented	DSC_0092_segment_1_segmented	1.000	0.366	4	0.224	0.512	0.051	0.0378
DSC_0093_segment_1_segmented	DSC_0093_segment_1_segmented	1.000	0.336	4	0.441	0.251	0.122	0.0322
DSC_0094_segment_1_segmented	DSC_0094_segment_1_segmented	1.000	0.456	4	0.094	0.326	0.079	0.0244
DSC_0096_segment_1_segmented	DSC_0096_segment_1_segmented	1.000	0.498	4	0.525	0.169	0.131	0.0132
DSC_0099_segment_1_segmented	DSC_0099_segment_1_segmented	1.000	0.448	4	0.373	0.050	0.167	0.0464
DSC_0100_segment_1_segmented	DSC_0100_segment_1_segmented	1.000	0.448	4	0.160	0.067	0.171	0.0337
DSC_0103_segment_1_segmented	DSC_0103_segment_1_segmented	1.000	0.336	4	0.286	0.486	0.128	0.0341
DSC_0105_segment_1_segmented	DSC_0105_segment_1_segmented	1.000	0.224	4	0.256	0.232	0.101	0.0011
DSC_0106_segment_1_segmented	DSC_0106_segment_1_segmented	1.000	0.336	4	0.077	0.310	0.162	0.0461
DSC_0107_segment_1_segmented	DSC_0107_segment_1_segmented	1.000	0.448	4	0.453	0.534	0.234	0.0315
DSC_0109_segment_1_segmented	DSC_0109_segment_1_segmented	1.000	0.448	4	0.050	0.121	0.180	0.0416
DSC_0110_segment_1_segmented	DSC_0110_segment_1_segmented	1.000	0.224	4	0.200	0.600	0.227	0.0500
DSC_0111_segment_1_segmented	DSC_0111_segment_1_segmented	1.000	0.336	4	0.461	0.079	0.236	0.0348
DSC_0112_segment_1_segmented	DSC_0112_segment_1_segmented	1.000	0.448	4	0.183	0.072	0.150	0.0500
DSC_0113_segment_1_segmented	DSC_0113_segment_1_segmented	1.000	0.336	4	0.316	0.079	0.328	0.0200
DSC_0114_segment_1_segmented	DSC_0114_segment_1_segmented	1.000	0.672	4	0.214	0.255	0.196	0.0081
DSC_0118_segment_1_segmented	DSC_0118_segment_1_segmented	1.000	0.336	4	0.453	0.478	0.203	0.0037
DSC_0119_segment_1_segmented	DSC_0119_segment_1_segmented	1.000	0.336	4	0.239	0.463	0.178	0.0183
DSC_0120_segment_1_segmented	DSC_0120_segment_1_segmented	1.000	0.336	4	0.050	0.430	0.204	0.0220
DSC_0121_segment_1_segmented	DSC_0121_segment_1_segmented	1.000	0.336	4	0.102	0.146	0.183	0.0233
DSC_0122_segment_1_segmented	DSC_0122_segment_1_segmented	1.000	0.448	4	0.257	0.526	0.129	0.0396
DSC_0127_segment_1_segmented	DSC_0127_segment_1_segmented	1.000	0.560	4	0.497	0.146	0.143	0.0157
DSC_0128_segment_1_segmented	DSC_0128_segment_1_segmented	1.000	0.448	4	0.182	0.549	0.239	0.0349
DSC_0129_segment_1_segmented	DSC_0129_segment_1_segmented	1.000	0.560	4	0.433	0.455	0.070	0.0299
DSC_0130_segment_1_segmented	DSC_0130_segment_1_segmented	1.000	0.336	4	0.062	0.191	0.135	0.0361
DSC_0132_segment_1_segmented	DSC_0132_segment_1_segmented	1.000	0.224	4	0.320	0.488	0.105	0.0252
DSC_0133_segment_1_segmented	DSC_0133_segment_1_segmented	1.000	0.336	4	0.337	0.600	0.098	0.0408
DSC_0135_segment_1_segmented	DSC_0135_segment_1_segmented	1.000	0.336	4	0.304	0.537	0.121	0.0104
DSC_0137_segment_1_segmented	DSC_0137_segment_1_segmented	1.000	0.448	4	0.398	0.305	0.093	0.0306
DSC_0138_segment_1_segmented	DSC_0138_segment_1_segmented	1.000	0.336	4	0.518	0.512	0.289	0.0341
DSC_0139_segment_1_segmented	DSC_0139_segment_1_segmented	1.000	0.448	4	0.323	0.270	0.227	0.0297
DSC_0140_segment_1_segmented	DSC_0140_segment_1_segmented	1.000	0.336	4	0.395	0.350	0.087	0.0494
DSC_0142_segment_1_segmented	DSC_0142_segment_1_segmented	1.000	0.448	4	0.596	0.098	0.213	0.0167
DSC_0146_segment_1_segmented	DSC_0146_segment_1_segmented	1.000	0.560	4	0.134	0.547	0.288	0.0092
DSC_0147_segment_1_segmented	DSC_0147_segment_1_segmented	1.000	0.224	4	0.321	0.082	0.190	0.0081
DSC_0148_segment_1_segmented	DSC_0148_segment_1_segmented	1.000	0.336	4	0.600	0.254	0.342	0.0166
DSC_0149_segment_1_segmented	DSC_0149_segment_1_segmented	1.000	0.448	4	0.136	0.099	0.101	0.0500
DSC_0154_segment_1_segmented	DSC_0154_segment_1_segmented	1.000	0.336	4	0.080	0.362	0.062	0.0068
DSC_0156_segment_1_segmented	DSC_0156_segment_1_segmented	1.000	0.542	4	0.543	0.246	0.208	0.0133
DSC_0157_segment_1_segmented	DSC_0157_segment_1_segmented	1.000	0.336	4	0.094	0.221	0.269	0.0091

Patch ID	Leaf ID	Score	Sec	Calls	μ	λ	Diff	α
DSC_0160_segment_1_segmented	DSC_0160_segment_1_segmented_smoothed	1.000	0.248	4	0.257	0.427	0.231	0.0449
DSC_0161_segment_1_segmented	DSC_0161_segment_1_segmented_smoothed	1.000	0.336	4	0.553	0.241	0.076	0.0281
DSC_0162_segment_1_segmented	DSC_0162_segment_1_segmented_smoothed	1.000	0.248	4	0.156	0.149	0.260	0.0188
DSC_0163_segment_1_segmented	DSC_0163_segment_1_segmented_smoothed	1.000	0.248	4	0.406	0.344	0.205	0.0045
DSC_0165_segment_1_segmented	DSC_0165_segment_1_segmented_smoothed	1.000	0.248	4	0.510	0.076	0.286	0.0358
DSC_0166_segment_1_segmented	DSC_0166_segment_1_segmented_smoothed	1.000	0.248	4	0.247	0.517	0.073	0.0350
DSC_0171_segment_1_segmented	DSC_0171_segment_1_segmented_smoothed	1.000	0.336	4	0.142	0.320	0.145	0.0453
DSC_0172_segment_1_segmented	DSC_0172_segment_1_segmented_smoothed	1.000	0.248	4	0.127	0.051	0.300	0.0268
DSC_0173_segment_1_segmented	DSC_0173_segment_1_segmented_smoothed	1.000	0.567	4	0.094	0.098	0.233	0.0253
DSC_0174_segment_1_segmented	DSC_0174_segment_1_segmented_smoothed	1.000	0.567	4	0.342	0.127	0.192	0.0500
DSC_0175_segment_1_segmented	DSC_0175_segment_1_segmented_smoothed	1.000	0.760	4	0.188	0.499	0.059	0.0057
DSC_0176_segment_1_segmented	DSC_0176_segment_1_segmented_smoothed	1.000	0.384	4	0.600	0.408	0.140	0.0160
DSC_0178_segment_1_segmented	DSC_0178_segment_1_segmented_smoothed	1.000	0.560	4	0.270	0.452	0.050	0.0031
DSC_0180_segment_1_segmented	DSC_0180_segment_1_segmented_smoothed	1.000	0.388	4	0.135	0.242	0.167	0.0105
DSC_0181_segment_1_segmented	DSC_0181_segment_1_segmented_smoothed	1.000	0.248	4	0.245	0.333	0.350	0.0070
DSC_0182_segment_1_segmented	DSC_0182_segment_1_segmented_smoothed	1.000	0.560	4	0.471	0.372	0.199	0.0482
DSC_0183_segment_1_segmented	DSC_0183_segment_1_segmented_smoothed	1.000	0.336	4	0.556	0.228	0.303	0.0421
DSC_0184_segment_1_segmented	DSC_0184_segment_1_segmented_smoothed	1.000	0.224	4	0.270	0.359	0.222	0.0441
DSC_0185_segment_1_segmented	DSC_0185_segment_1_segmented_smoothed	1.000	0.552	4	0.153	0.150	0.239	0.0306
DSC_0186_segment_1_segmented	DSC_0186_segment_1_segmented_smoothed	1.000	0.464	4	0.468	0.109	0.080	0.0094
DSC_0187_segment_1_segmented	DSC_0187_segment_1_segmented_smoothed	1.000	0.224	4	0.089	0.225	0.149	0.0452
DSC_0188_segment_1_segmented	DSC_0188_segment_1_segmented_smoothed	1.000	0.648	4	0.075	0.162	0.074	0.0422
DSC_0191_segment_1_segmented	DSC_0191_segment_1_segmented_smoothed	1.000	0.560	4	0.136	0.438	0.078	0.0142
DSC_0192_segment_1_segmented	DSC_0192_segment_1_segmented_smoothed	1.000	0.760	4	0.292	0.310	0.102	0.0109
DSC_0198_segment_1_segmented	DSC_0198_segment_1_segmented_smoothed	1.000	1.408	4	0.600	0.350	0.246	0.0349
DSC_0199_segment_1_segmented	DSC_0199_segment_1_segmented_smoothed	1.000	0.567	4	0.072	0.221	0.241	0.0098
DSC_0201_segment_1_segmented	DSC_0201_segment_1_segmented_smoothed	1.000	0.648	4	0.430	0.456	0.299	0.0318
DSC_0203_segment_1_segmented	DSC_0203_segment_1_segmented_smoothed	1.000	0.424	4	0.215	0.537	0.093	0.0286
DSC_0205_segment_1_segmented	DSC_0205_segment_1_segmented_smoothed	1.000	0.560	4	0.113	0.209	0.158	0.0111
DSC_0206_segment_1_segmented	DSC_0206_segment_1_segmented_smoothed	1.000	0.648	4	0.050	0.155	0.255	0.0389
DSC_0208_segment_1_segmented	DSC_0208_segment_1_segmented_smoothed	1.000	0.336	4	0.428	0.140	0.183	0.0485
DSC_0209_segment_1_segmented	DSC_0209_segment_1_segmented_smoothed	1.000	0.336	4	0.206	0.414	0.204	0.0216
DSC_0210_segment_1_segmented	DSC_0210_segment_1_segmented_smoothed	1.000	0.248	4	0.449	0.173	0.114	0.0248
DSC_0211_segment_1_segmented	DSC_0211_segment_1_segmented_smoothed	1.000	0.728	4	0.332	0.374	0.074	0.0210
DSC_0213_segment_1_segmented	DSC_0213_segment_1_segmented_smoothed	1.000	0.372	4	0.215	0.439	0.178	0.0445
DSC_0217_segment_1_segmented	DSC_0217_segment_1_segmented_smoothed	1.000	0.336	4	0.190	0.156	0.184	0.0421
DSC_0218_segment_1_segmented	DSC_0218_segment_1_segmented_smoothed	1.000	0.896	4	0.238	0.133	0.201	0.0438
DSC_0220_segment_1_segmented	DSC_0220_segment_1_segmented_smoothed	1.000	0.432	4	0.130	0.419	0.132	0.0144
DSC_0222_segment_1_segmented	DSC_0222_segment_1_segmented_smoothed	1.000	0.536	4	0.417	0.239	0.115	0.0142
DSC_0223_segment_1_segmented	DSC_0223_segment_1_segmented_smoothed	1.000	0.784	4	0.371	0.094	0.104	0.0294
DSC_0225_segment_1_segmented	DSC_0225_segment_1_segmented_smoothed	1.000	0.552	4	0.068	0.544	0.288	0.0069
DSC_0226_segment_1_segmented	DSC_0226_segment_1_segmented_smoothed	1.000	0.424	4	0.234	0.322	0.229	0.0136
DSC_0227_segment_1_segmented	DSC_0227_segment_1_segmented_smoothed	1.000	0.336	4	0.203	0.338	0.332	0.0075
DSC_0232_segment_1_segmented	DSC_0232_segment_1_segmented_smoothed	1.000	0.536	4	0.132	0.148	0.271	0.0462
DSC_0235_segment_1_segmented	DSC_0235_segment_1_segmented_smoothed	1.000	0.560	4	0.281	0.096	0.156	0.0246
DSC_0239_segment_1_segmented	DSC_0239_segment_1_segmented_smoothed	1.000	0.336	4	0.600	0.273	0.115	0.0182

Patch ID	Leaf ID	Score	Sec	Calls	μ	λ	Diff	α
DSC_0240_segment_1_segmented	DSC_0240_segment_1_segmented	1.000	0.000	4	0.148	0.427	0.094	0.0372
DSC_0241_segment_1_segmented	DSC_0241_segment_1_segmented	1.000	0.000	4	0.200	0.147	0.129	0.0044
DSC_0242_segment_1_segmented	DSC_0242_segment_1_segmented	1.000	0.000	4	0.384	0.559	0.126	0.0309
DSC_0243_segment_1_segmented	DSC_0243_segment_1_segmented	1.000	0.000	4	0.349	0.173	0.234	0.0140
DSC_0248_segment_1_segmented	DSC_0248_segment_1_segmented	1.000	0.000	4	0.118	0.135	0.125	0.0060
DSC_0249_segment_1_segmented	DSC_0249_segment_1_segmented	1.000	0.000	4	0.148	0.445	0.132	0.0169
DSC_0251_segment_1_segmented	DSC_0251_segment_1_segmented	1.000	0.000	4	0.496	0.244	0.166	0.0294
DSC_0251_segment_1_segmented	DSC_0251_segment_1_segmented_ps224_segmented	0.611	0.000	4	0.101	0.050	0.305	0.0266
DSC_0252_segment_1_segmented	DSC_0252_segment_1_segmented	1.000	0.000	4	0.078	0.217	0.310	0.0403
DSC_0252_segment_1_segmented	DSC_0252_segment_1_segmented_ps224_segmented	0.711	0.000	4	0.275	0.361	0.147	0.0144
DSC_0253_segment_1_segmented	DSC_0253_segment_1_segmented	1.000	0.000	4	0.071	0.389	0.123	0.0347
DSC_0254_segment_1_segmented	DSC_0254_segment_1_segmented	1.000	0.000	4	0.173	0.350	0.078	0.0238
DSC_0256_segment_1_segmented	DSC_0256_segment_1_segmented	1.000	0.000	4	0.332	0.284	0.208	0.0212
DSC_0256_segment_1_segmented	DSC_0256_segment_1_segmented_ps224_segmented	0.489	0.000	4	0.123	0.368	0.156	0.0194
DSC_0257_segment_1_segmented	DSC_0257_segment_1_segmented	1.000	0.000	4	0.345	0.329	0.160	0.0107
DSC_0257_segment_1_segmented	DSC_0257_segment_1_segmented_ps224_segmented	0.618	0.000	4	0.056	0.382	0.132	0.0261
DSC_0258_segment_1_segmented	DSC_0258_segment_1_segmented	1.000	0.000	4	0.185	0.454	0.254	0.0084
DSC_0260_segment_1_segmented	DSC_0260_segment_1_segmented	1.000	0.000	4	0.553	0.051	0.253	0.0459
DSC_0260_segment_1_segmented	DSC_0260_segment_1_segmented_ps224_segmented	0.638	0.000	4	0.488	0.515	0.050	0.0209
DSC_0261_segment_1_segmented	DSC_0261_segment_1_segmented	1.000	0.000	4	0.577	0.375	0.118	0.0500
DSC_0262_segment_1_segmented	DSC_0262_segment_1_segmented	1.000	0.000	4	0.244	0.105	0.216	0.0433
DSC_0263_segment_1_segmented	DSC_0263_segment_1_segmented	1.000	0.000	4	0.600	0.579	0.060	0.0362
DSC_0264_segment_1_segmented	DSC_0264_segment_1_segmented	1.000	0.000	4	0.050	0.278	0.296	0.0204
DSC_0265_segment_1_segmented	DSC_0265_segment_1_segmented	1.000	0.000	4	0.447	0.548	0.111	0.0051
DSC_0267_segment_1_segmented	DSC_0267_segment_1_segmented	1.000	0.000	4	0.600	0.272	0.107	0.0152
DSC_0268_segment_1_segmented	DSC_0268_segment_1_segmented	1.000	0.000	4	0.493	0.242	0.106	0.0437
DSC_0269_segment_1_segmented	DSC_0269_segment_1_segmented	1.000	0.000	4	0.341	0.595	0.218	0.0319
DSC_0270_segment_1_segmented	DSC_0270_segment_1_segmented	1.000	0.000	4	0.173	0.396	0.152	0.0172
DSC_0272_segment_1_segmented	DSC_0272_segment_1_segmented	1.000	0.000	4	0.505	0.305	0.143	0.0029
DSC_0273_segment_1_segmented	DSC_0273_segment_1_segmented	1.000	0.000	4	0.106	0.206	0.159	0.0394
DSC_0274_segment_1_segmented	DSC_0274_segment_1_segmented	1.000	0.000	4	0.050	0.600	0.314	0.0383
DSC_0275_segment_1_segmented	DSC_0275_segment_1_segmented	1.000	0.000	4	0.176	0.050	0.312	0.0429
DSC_0277_segment_1_segmented	DSC_0277_segment_1_segmented	1.000	0.000	4	0.181	0.593	0.119	0.0202
DSC_0278_segment_1_segmented	DSC_0278_segment_1_segmented	1.000	0.000	4	0.182	0.562	0.116	0.0339
DSC_0279_segment_1_segmented	DSC_0279_segment_1_segmented	1.000	0.000	4	0.265	0.553	0.126	0.0122
DSC_0283_segment_1_segmented	DSC_0283_segment_1_segmented	1.000	0.000	4	0.469	0.131	0.232	0.0489
DSC_0284_segment_1_segmented	DSC_0284_segment_1_segmented	1.000	0.000	4	0.464	0.201	0.125	0.0212
DSC_0285_segment_1_segmented	DSC_0285_segment_1_segmented	1.000	0.000	4	0.139	0.342	0.245	0.0274
DSC_0286_segment_1_segmented	DSC_0286_segment_1_segmented	1.000	0.000	4	0.217	0.085	0.070	0.0017
DSC_0287_segment_1_segmented	DSC_0287_segment_1_segmented	1.000	0.000	4	0.217	0.275	0.257	0.0228
DSC_0288_segment_1_segmented	DSC_0288_segment_1_segmented	1.000	0.000	4	0.262	0.050	0.082	0.0372
DSC_0363_segment_1_segmented	DSC_0363_segment_1_segmented	1.000	0.000	4	0.192	0.174	0.050	0.0103
DSC_0364_segment_1_segmented	DSC_0364_segment_1_segmented	1.000	0.000	4	0.517	0.050	0.260	0.0305
DSC_0365_segment_1_segmented	DSC_0365_segment_1_segmented	1.000	0.000	4	0.457	0.345	0.120	0.0255
DSC_0370_segment_1_segmented	DSC_0370_segment_1_segmented	1.000	0.000	4	0.050	0.479	0.169	0.0121
DSC_0371_segment_1_segmented	DSC_0371_segment_1_segmented	1.000	0.000	4	0.372	0.299	0.173	0.0445

Patch ID	Leaf ID	Score	Sec	Calls	μ	λ	Diff	α
DSC_0372_segment_1_segmented	DSC_0372_segment_1_segmented	112.3868	0.3468	4	0.122	0.262	0.303	0.0372
DSC_0374_segment_1_segmented	DSC_0374_segment_1_segmented	112.6596	0.4668	4	0.105	0.352	0.316	0.0152
DSC_0376_segment_1_segmented	DSC_0376_segment_1_segmented	112.5624	0.3567	4	0.168	0.196	0.061	0.0077
DSC_0378_segment_1_segmented	DSC_0378_segment_1_segmented	112.8377	0.4360	4	0.050	0.327	0.116	0.0238
DSC_0379_segment_1_segmented	DSC_0379_segment_1_segmented	112.7367	0.2548	4	0.320	0.154	0.329	0.0285
DSC_0380_segment_1_segmented	DSC_0380_segment_1_segmented	112.0928	0.2760	4	0.135	0.321	0.239	0.0393
DSC_0381_segment_1_segmented	DSC_0381_segment_1_segmented	112.0425	0.4650	4	0.494	0.382	0.144	0.0170
DSC_0386_segment_1_segmented	DSC_0386_segment_1_segmented	112.6123	0.4478	4	0.476	0.108	0.071	0.0200
DSC_0389_segment_1_segmented	DSC_0389_segment_1_segmented	112.2635	0.8048	4	0.050	0.359	0.243	0.0340
DSC_0394_segment_1_segmented	DSC_0394_segment_1_segmented	112.2832	0.2760	4	0.444	0.454	0.089	0.0105
DSC_0397_segment_1_segmented	DSC_0397_segment_1_segmented	112.2868	0.3452	4	0.525	0.366	0.053	0.0384
DSC_0398_segment_1_segmented	DSC_0398_segment_1_segmented	112.2536	0.3760	4	0.554	0.415	0.296	0.0399
DSC_0400_segment_1_segmented	DSC_0400_segment_1_segmented	112.1486	0.2760	4	0.190	0.478	0.050	0.0258
DSC_0401_segment_1_segmented	DSC_0401_segment_1_segmented	112.3724	0.3560	4	0.286	0.101	0.207	0.0412
DSC_0402_segment_1_segmented	DSC_0402_segment_1_segmented	112.6700	0.2468	4	0.467	0.245	0.179	0.0081
DSC_0404_segment_1_segmented	DSC_0404_segment_1_segmented	112.7356	0.4896	4	0.053	0.237	0.265	0.0377
DSC_0406_segment_1_segmented	DSC_0406_segment_1_segmented	112.1501	0.5760	4	0.442	0.441	0.132	0.0421
DSC_0407_segment_1_segmented	DSC_0407_segment_1_segmented	112.0126	0.2768	4	0.436	0.582	0.171	0.0184
DSC_0408_segment_1_segmented	DSC_0408_segment_1_segmented	112.0015	0.6848	4	0.139	0.506	0.350	0.0091
DSC_0412_segment_1_segmented	DSC_0412_segment_1_segmented	112.8736	0.4560	4	0.285	0.131	0.187	0.0287
DSC_0416_segment_1_segmented	DSC_0416_segment_1_segmented	112.0924	0.5568	4	0.050	0.092	0.198	0.0414
DSC_0417_segment_1_segmented	DSC_0417_segment_1_segmented	112.8268	0.2760	4	0.149	0.134	0.155	0.0371
DSC_0419_segment_1_segmented	DSC_0419_segment_1_segmented	112.0736	0.1560	4	0.362	0.393	0.350	0.0459

14.4 Casewise Patch Prediction Records

Table 11: Casewise patch prediction and error records on primary leaves.

Patch ID	Leaf ID	Split	Unc	AbsErr	μ^*	$\hat{\mu}$	λ^*	$\hat{\lambda}$
DSC_0024_segment_1_segmented	DSC_0024_segment_1_segmented	112	224.0336	1.891	0.105	0.685	0.556	0.010
DSC_0036_segment_1_segmented	DSC_0036_segment_1_segmented	112	322.6672	6.339	0.536	0.358	0.501	-0.372
DSC_0044_segment_1_segmented	DSC_0044_segment_1_segmented	112	112.6672	6.608	0.194	0.600	0.096	-0.167
DSC_0050_segment_1_segmented	DSC_0050_segment_1_segmented	112	672.9656	5.492	0.480	0.465	0.463	-0.365
DSC_0070_segment_1_segmented	DSC_0070_segment_1_segmented	112	112.4776	6.173	0.107	0.638	0.050	-0.010
DSC_0094_segment_1_segmented	DSC_0094_segment_1_segmented	112	201.6456	0.861	0.094	0.518	0.326	0.038
DSC_0107_segment_1_segmented	DSC_0107_segment_1_segmented	112	280.7248	2.033	0.453	0.305	0.534	0.217
DSC_0109_segment_1_segmented	DSC_0109_segment_1_segmented	112	366.6448	3.306	0.050	0.149	0.121	-0.014
DSC_0120_segment_1_segmented	DSC_0120_segment_1_segmented	112	170.2836	1.718	0.050	0.227	0.430	-0.039
DSC_0128_segment_1_segmented	DSC_0128_segment_1_segmented	112	346.6948	1.175	0.182	0.177	0.549	0.027
DSC_0130_segment_1_segmented	DSC_0130_segment_1_segmented	112	342.3836	7.696	0.062	0.187	0.191	0.045
DSC_0148_segment_1_segmented	DSC_0148_segment_1_segmented	112	102.6236	3.193	0.600	0.203	0.254	0.129
DSC_0163_segment_1_segmented	DSC_0163_segment_1_segmented	112	201.2548	7.284	0.406	0.229	0.344	0.009
DSC_0173_segment_1_segmented	DSC_0173_segment_1_segmented	112	367.6560	1.476	0.094	0.462	0.098	-0.011
DSC_0175_segment_1_segmented	DSC_0175_segment_1_segmented	112	201.2560	0.197	0.188	0.268	0.499	-0.077
DSC_0184_segment_1_segmented	DSC_0184_segment_1_segmented	112	156.8224	3.354	0.270	0.352	0.359	0.046
DSC_0187_segment_1_segmented	DSC_0187_segment_1_segmented	112	318.8224	3.289	0.089	0.173	0.225	0.038

Patch ID	Leaf ID	Split	Unc	AbsErr	μ^*	$\hat{\mu}$	λ^*	$\hat{\lambda}$
DSC_0188_segment_1_segmented	DSC_0188_segment_1_segmented	224_112_2	0.280	0.280	0.075	0.323	0.162	0.153
DSC_0191_segment_1_segmented	DSC_0191_segment_1_segmented	224_112_2	2.623	2.623	0.136	0.300	0.438	0.102
DSC_0199_segment_1_segmented	DSC_0199_segment_1_segmented	224_112_0	1.168	1.168	0.072	0.266	0.221	0.431
DSC_0205_segment_1_segmented	DSC_0205_segment_1_segmented	224_112_2	1.832	1.832	0.113	0.150	0.209	0.048
DSC_0210_segment_1_segmented	DSC_0210_segment_1_segmented	224_112_3	1.587	1.587	0.449	0.121	0.173	0.101
DSC_0223_segment_1_segmented	DSC_0223_segment_1_segmented	224_112_6	6.265	6.265	0.371	0.315	0.094	0.236
DSC_0235_segment_1_segmented	DSC_0235_segment_1_segmented	224_112_2	6.107	6.107	0.281	0.280	0.096	0.070
DSC_0239_segment_1_segmented	DSC_0239_segment_1_segmented	224_112_3	7.356	7.356	0.600	0.127	0.273	0.067
DSC_0240_segment_1_segmented	DSC_0240_segment_1_segmented	224_112_3	2.202	2.202	0.148	0.151	0.427	0.103
DSC_0248_segment_1_segmented	DSC_0248_segment_1_segmented	224_112_6	3.723	3.723	0.118	0.414	0.135	-0.391
DSC_0257_segment_1_segmented	DSC_0257_segment_1_segmented	224_112_5	2.375	2.375	0.345	0.185	0.329	0.537
DSC_0257_segment_1_segmented	DSC_0257_segment_1_segmented	224_112_2	48.529.1	48.529.1	0.056	0.445	0.382	0.039
DSC_0262_segment_1_segmented	DSC_0262_segment_1_segmented	224_112_2	7.390	7.390	0.244	0.204	0.105	0.057
DSC_0267_segment_1_segmented	DSC_0267_segment_1_segmented	224_112_1	5.401	5.401	0.600	0.296	0.272	0.073
DSC_0269_segment_1_segmented	DSC_0269_segment_1_segmented	224_112_2	1.683	1.683	0.341	0.330	0.595	0.145
DSC_0279_segment_1_segmented	DSC_0279_segment_1_segmented	224_112_2	0.860	0.860	0.265	0.325	0.553	0.268
DSC_0284_segment_1_segmented	DSC_0284_segment_1_segmented	224_112_3	5.602	5.602	0.464	0.747	0.201	0.382
DSC_0394_segment_1_segmented	DSC_0394_segment_1_segmented	224_112_3	6.466	6.466	0.444	0.212	0.454	0.251
DSC_0018_segment_1_segmented	DSC_0018_segment_1_segmented	224_112_4	7.368	7.368	0.050	0.513	0.271	-0.019
DSC_0034_segment_1_segmented	DSC_0034_segment_1_segmented	224_112_3	2.359	2.359	0.083	0.614	0.436	-0.116
DSC_0037_segment_1_segmented	DSC_0037_segment_1_segmented	224_112_1	0.114	0.114	0.127	0.616	0.050	0.011
DSC_0040_segment_1_segmented	DSC_0040_segment_1_segmented	224_112_5	6.533	6.533	0.106	0.541	0.238	-0.135
DSC_0059_segment_1_segmented	DSC_0059_segment_1_segmented	224_112_4	3.053	3.053	0.180	0.447	0.472	-0.025
DSC_0068_segment_1_segmented	DSC_0068_segment_1_segmented	224_112_0	3.266	3.266	0.355	0.587	0.505	-0.002
DSC_0069_segment_1_segmented	DSC_0069_segment_1_segmented	224_112_4	5.535	5.535	0.110	0.466	0.139	-0.157
DSC_0084_segment_1_segmented	DSC_0084_segment_1_segmented	224_112_1	1.729	1.729	0.217	0.509	0.178	-0.300
DSC_0090_segment_1_segmented	DSC_0090_segment_1_segmented	224_112_2	7.532	7.532	0.073	0.421	0.050	-0.015
DSC_0091_segment_1_segmented	DSC_0091_segment_1_segmented	224_112_0	1.097	1.097	0.083	0.262	0.198	-0.055
DSC_0092_segment_1_segmented	DSC_0092_segment_1_segmented	224_112_3	6.766	6.766	0.224	0.537	0.512	-0.035
DSC_0093_segment_1_segmented	DSC_0093_segment_1_segmented	224_112_1	7.518	7.518	0.441	-0.041	0.251	0.068
DSC_0096_segment_1_segmented	DSC_0096_segment_1_segmented	224_112_3	5.695	5.695	0.525	0.375	0.169	0.305
DSC_0100_segment_1_segmented	DSC_0100_segment_1_segmented	224_112_2	3.185	3.185	0.160	0.415	0.067	0.126
DSC_0103_segment_1_segmented	DSC_0103_segment_1_segmented	224_112_3	5.509	5.509	0.286	0.181	0.486	-0.012
DSC_0105_segment_1_segmented	DSC_0105_segment_1_segmented	224_112_3	6.883	6.883	0.256	0.227	0.232	0.134
DSC_0106_segment_1_segmented	DSC_0106_segment_1_segmented	224_112_1	7.211	7.211	0.077	0.223	0.310	0.039
DSC_0110_segment_1_segmented	DSC_0110_segment_1_segmented	224_112_3	6.051	6.051	0.200	0.575	0.600	0.123
DSC_0113_segment_1_segmented	DSC_0113_segment_1_segmented	224_112_3	6.481	6.481	0.316	0.129	0.079	0.077
DSC_0114_segment_1_segmented	DSC_0114_segment_1_segmented	224_112_3	1.002	1.002	0.214	0.474	0.255	0.281
DSC_0119_segment_1_segmented	DSC_0119_segment_1_segmented	224_112_3	0.775	0.775	0.239	0.225	0.463	-0.021
DSC_0121_segment_1_segmented	DSC_0121_segment_1_segmented	224_112_2	1.748	1.748	0.102	0.283	0.146	0.060
DSC_0122_segment_1_segmented	DSC_0122_segment_1_segmented	224_112_2	7.499	7.499	0.257	0.296	0.526	0.001
DSC_0127_segment_1_segmented	DSC_0127_segment_1_segmented	224_112_3	5.094	5.094	0.497	0.377	0.146	0.053
DSC_0129_segment_1_segmented	DSC_0129_segment_1_segmented	224_112_3	6.414	6.414	0.433	0.272	0.455	0.084
DSC_0132_segment_1_segmented	DSC_0132_segment_1_segmented	224_112_2	1.709	1.709	0.320	0.158	0.488	0.123
DSC_0133_segment_1_segmented	DSC_0133_segment_1_segmented	224_112_1	2.608	2.608	0.337	0.192	0.600	-0.020
DSC_0137_segment_1_segmented	DSC_0137_segment_1_segmented	224_112_2	7.182	7.182	0.398	0.190	0.305	0.128

Patch ID	Leaf ID	Split	Unc	AbsErr	μ^*	$\hat{\mu}$	λ^*	$\hat{\lambda}$
DSC_0138_segment_1	DSC_0138_segment_1_segmented	[illegible]	[illegible]	2.667	0.518	0.119	0.512	-0.069
DSC_0139_segment_1	DSC_0139_segment_1_segmented	[illegible]	[illegible]	7.636	0.323	0.158	0.270	0.012
DSC_0140_segment_1	DSC_0140_segment_1_segmented	[illegible]	[illegible]	5.534	0.395	0.493	0.350	0.535
DSC_0147_segment_1	DSC_0147_segment_1_segmented	[illegible]	[illegible]	1.158	0.321	0.322	0.082	0.012
DSC_0149_segment_1	DSC_0149_segment_1_segmented	[illegible]	[illegible]	5.420	0.136	0.279	0.099	0.231
DSC_0157_segment_1	DSC_0157_segment_1_segmented	[illegible]	[illegible]	0.152	0.094	0.131	0.221	-0.009
DSC_0160_segment_1	DSC_0160_segment_1_segmented	[illegible]	[illegible]	0.358	0.257	0.132	0.427	0.085
DSC_0162_segment_1	DSC_0162_segment_1_segmented	[illegible]	[illegible]	3.084	0.156	0.375	0.149	-0.232
DSC_0165_segment_1	DSC_0165_segment_1_segmented	[illegible]	[illegible]	0.409	0.510	0.469	0.076	0.487
DSC_0172_segment_1	DSC_0172_segment_1_segmented	[illegible]	[illegible]	7.140	0.127	0.163	0.051	-0.041
DSC_0174_segment_1	DSC_0174_segment_1_segmented	[illegible]	[illegible]	1.065	0.342	0.388	0.127	-0.002
DSC_0176_segment_1	DSC_0176_segment_1_segmented	[illegible]	[illegible]	6.705	0.600	0.202	0.408	0.375
DSC_0178_segment_1	DSC_0178_segment_1_segmented	[illegible]	[illegible]	7.333	0.270	0.293	0.452	0.087
DSC_0180_segment_1	DSC_0180_segment_1_segmented	[illegible]	[illegible]	3.442	0.135	0.366	0.242	0.316
DSC_0181_segment_1	DSC_0181_segment_1_segmented	[illegible]	[illegible]	0.757	0.245	0.152	0.333	0.108
DSC_0182_segment_1	DSC_0182_segment_1_segmented	[illegible]	[illegible]	5.904	0.471	0.482	0.372	-0.168
DSC_0185_segment_1	DSC_0185_segment_1_segmented	[illegible]	[illegible]	1.288	0.153	0.442	0.150	-0.187
DSC_0186_segment_1	DSC_0186_segment_1_segmented	[illegible]	[illegible]	1.034	0.468	0.393	0.109	0.137
DSC_0203_segment_1	DSC_0203_segment_1_segmented	[illegible]	[illegible]	1.222	0.215	0.414	0.537	-0.006
DSC_0208_segment_1	DSC_0208_segment_1_segmented	[illegible]	[illegible]	2.565	0.428	0.265	0.140	0.158
DSC_0209_segment_1	DSC_0209_segment_1_segmented	[illegible]	[illegible]	3.055	0.206	0.422	0.414	0.327
DSC_0211_segment_1	DSC_0211_segment_1_segmented	[illegible]	[illegible]	1.550	0.332	0.524	0.374	0.301
DSC_0213_segment_1	DSC_0213_segment_1_segmented	[illegible]	[illegible]	1.156	0.215	0.271	0.439	0.094
DSC_0218_segment_1	DSC_0218_segment_1_segmented	[illegible]	[illegible]	0.669	0.238	0.364	0.133	0.457
DSC_0220_segment_1	DSC_0220_segment_1_segmented	[illegible]	[illegible]	1.315	0.130	0.355	0.419	-0.115
DSC_0222_segment_1	DSC_0222_segment_1_segmented	[illegible]	[illegible]	0.174	0.417	0.288	0.239	0.109
DSC_0225_segment_1	DSC_0225_segment_1_segmented	[illegible]	[illegible]	1.395	0.068	0.383	0.544	0.004
DSC_0226_segment_1	DSC_0226_segment_1_segmented	[illegible]	[illegible]	7.353	0.234	0.162	0.322	0.042
DSC_0227_segment_1	DSC_0227_segment_1_segmented	[illegible]	[illegible]	5.140	0.203	0.300	0.338	-0.011
DSC_0232_segment_1	DSC_0232_segment_1_segmented	[illegible]	[illegible]	6.485	0.132	0.186	0.148	0.076
DSC_0242_segment_1	DSC_0242_segment_1_segmented	[illegible]	[illegible]	1.072	0.384	0.335	0.559	0.031
DSC_0243_segment_1	DSC_0243_segment_1_segmented	[illegible]	[illegible]	5.238	0.349	0.368	0.173	0.092
DSC_0249_segment_1	DSC_0249_segment_1_segmented	[illegible]	[illegible]	1.008	0.148	0.372	0.445	0.011
DSC_0251_segment_1	DSC_0251_segment_1_segmented	[illegible]	[illegible]	367.1	0.101	0.324	0.050	-0.216
DSC_0252_segment_1	DSC_0252_segment_1_segmented	[illegible]	[illegible]	1.409	0.078	0.110	0.217	0.056
DSC_0252_segment_1	DSC_0252_segment_1_segmented	[illegible]	[illegible]	366.1	0.275	0.455	0.361	0.003
DSC_0253_segment_1	DSC_0253_segment_1_segmented	[illegible]	[illegible]	0.273	0.071	0.501	0.389	0.203
DSC_0254_segment_1	DSC_0254_segment_1_segmented	[illegible]	[illegible]	5.280	0.173	0.176	0.350	-0.044
DSC_0256_segment_1	DSC_0256_segment_1_segmented	[illegible]	[illegible]	1.714	0.332	0.354	0.284	0.405
DSC_0256_segment_1	DSC_0256_segment_1_segmented	[illegible]	[illegible]	214.1	0.123	0.281	0.368	-0.079
DSC_0260_segment_1	DSC_0260_segment_1_segmented	[illegible]	[illegible]	5.433	0.553	0.255	0.051	0.064
DSC_0260_segment_1	DSC_0260_segment_1_segmented	[illegible]	[illegible]	5.837.1	0.488	0.386	0.515	-0.254
DSC_0261_segment_1	DSC_0261_segment_1_segmented	[illegible]	[illegible]	2.040	0.577	0.422	0.375	-0.161
DSC_0264_segment_1	DSC_0264_segment_1_segmented	[illegible]	[illegible]	2.307	0.050	0.215	0.278	0.033
DSC_0265_segment_1	DSC_0265_segment_1_segmented	[illegible]	[illegible]	1.061	0.447	0.434	0.548	0.025
DSC_0270_segment_1	DSC_0270_segment_1_segmented	[illegible]	[illegible]	5.722	0.173	0.503	0.396	-0.043

Patch ID	Leaf ID	Split	Unc	AbsErr	μ^*	$\hat{\mu}$	λ^*	$\hat{\lambda}$
DSC_0272_segment_1_segmented	DSC_0272_segment_224_segmented	train	0.336	0.030	0.505	0.093	0.305	0.082
DSC_0274_segment_1_segmented	DSC_0274_segment_224_segmented	train	0.560	5.663	0.050	0.396	0.600	0.104
DSC_0275_segment_1_segmented	DSC_0275_segment_224_segmented	train	0.448	6.723	0.176	0.140	0.050	0.003
DSC_0277_segment_1_segmented	DSC_0277_segment_224_segmented	train	0.448	5.287	0.181	0.216	0.593	-0.044
DSC_0283_segment_1_segmented	DSC_0283_segment_224_segmented	train	0.560	0.515	0.469	0.374	0.131	-0.026
DSC_0285_segment_1_segmented	DSC_0285_segment_224_segmented	train	0.560	6.052	0.139	0.461	0.342	-0.137
DSC_0286_segment_1_segmented	DSC_0286_segment_224_segmented	train	0.224	0.139	0.217	0.311	0.085	-0.060
DSC_0364_segment_1_segmented	DSC_0364_segment_224_segmented	train	0.448	7.013	0.517	0.134	0.050	0.084
DSC_0365_segment_1_segmented	DSC_0365_segment_224_segmented	train	0.448	0.794	0.457	0.317	0.345	0.185
DSC_0370_segment_1_segmented	DSC_0370_segment_224_segmented	train	0.560	3.710	0.050	0.195	0.479	0.185
DSC_0372_segment_1_segmented	DSC_0372_segment_224_segmented	train	0.448	6.015	0.122	0.214	0.262	0.160
DSC_0374_segment_1_segmented	DSC_0374_segment_224_segmented	train	0.448	6.440	0.105	0.340	0.352	0.010
DSC_0376_segment_1_segmented	DSC_0376_segment_224_segmented	train	0.560	0.792	0.168	0.179	0.196	-0.014
DSC_0379_segment_1_segmented	DSC_0379_segment_224_segmented	train	0.448	5.235	0.320	0.305	0.154	0.025
DSC_0380_segment_1_segmented	DSC_0380_segment_224_segmented	train	0.560	1.600	0.135	0.202	0.321	0.023
DSC_0381_segment_1_segmented	DSC_0381_segment_224_segmented	train	0.560	1.410	0.494	0.572	0.382	0.098
DSC_0386_segment_1_segmented	DSC_0386_segment_224_segmented	train	0.448	1.705	0.476	0.505	0.108	0.048
DSC_0389_segment_1_segmented	DSC_0389_segment_224_segmented	train	0.448	0.278	0.050	0.655	0.359	0.030
DSC_0397_segment_1_segmented	DSC_0397_segment_224_segmented	train	0.412	6.011	0.525	0.467	0.366	0.602
DSC_0398_segment_1_segmented	DSC_0398_segment_224_segmented	train	0.560	1.354	0.554	0.216	0.415	0.237
DSC_0400_segment_1_segmented	DSC_0400_segment_224_segmented	train	0.560	7.307	0.190	0.435	0.478	0.160
DSC_0401_segment_1_segmented	DSC_0401_segment_224_segmented	train	0.560	1.485	0.286	0.378	0.101	0.062
DSC_0402_segment_1_segmented	DSC_0402_segment_224_segmented	train	0.448	1.580	0.467	0.191	0.245	-0.093
DSC_0404_segment_1_segmented	DSC_0404_segment_224_segmented	train	0.896	2.035	0.053	0.279	0.237	0.388
DSC_0406_segment_1_segmented	DSC_0406_segment_224_segmented	train	0.560	7.073	0.442	0.380	0.441	0.215
DSC_0408_segment_1_segmented	DSC_0408_segment_224_segmented	train	0.448	1.125	0.139	0.361	0.506	0.262
DSC_0412_segment_1_segmented	DSC_0412_segment_224_segmented	train	0.560	2.483	0.285	0.461	0.131	-0.000
DSC_0416_segment_1_segmented	DSC_0416_segment_224_segmented	train	0.560	3.415	0.050	0.643	0.092	0.007
DSC_0417_segment_1_segmented	DSC_0417_segment_224_segmented	train	0.560	5.455	0.149	0.182	0.134	-0.024
DSC_0419_segment_1_segmented	DSC_0419_segment_224_segmented	train	0.560	1.149	0.362	0.313	0.393	0.024
DSC_0057_segment_1_segmented	DSC_0057_segment_224_segmented	val	0.448	3.188	0.243	0.532	0.352	-0.170
DSC_0060_segment_1_segmented	DSC_0060_segment_224_segmented	val	0.560	7.386	0.358	0.694	0.387	0.082
DSC_0064_segment_1_segmented	DSC_0064_segment_224_segmented	val	0.560	5.299	0.060	0.576	0.232	-0.109
DSC_0089_segment_1_segmented	DSC_0089_segment_224_segmented	val	0.560	0.637	0.074	0.602	0.509	-0.286
DSC_0099_segment_1_segmented	DSC_0099_segment_224_segmented	val	0.448	7.811	0.373	0.171	0.050	0.019
DSC_0111_segment_1_segmented	DSC_0111_segment_224_segmented	val	0.336	0.021	0.461	0.242	0.079	0.035
DSC_0112_segment_1_segmented	DSC_0112_segment_224_segmented	val	0.448	2.001	0.183	0.142	0.072	0.079
DSC_0118_segment_1_segmented	DSC_0118_segment_224_segmented	val	0.336	7.113	0.453	0.493	0.478	0.127
DSC_0135_segment_1_segmented	DSC_0135_segment_224_segmented	val	0.336	6.315	0.304	0.396	0.537	-0.056
DSC_0142_segment_1_segmented	DSC_0142_segment_224_segmented	val	0.448	0.235	0.596	0.068	0.098	0.214
DSC_0146_segment_1_segmented	DSC_0146_segment_224_segmented	val	0.560	1.360	0.134	0.574	0.547	0.744
DSC_0154_segment_1_segmented	DSC_0154_segment_224_segmented	val	0.336	6.393	0.080	0.245	0.362	0.168
DSC_0156_segment_1_segmented	DSC_0156_segment_224_segmented	val	0.542	1.069	0.543	0.294	0.246	0.483
DSC_0161_segment_1_segmented	DSC_0161_segment_224_segmented	val	0.336	5.399	0.553	0.208	0.241	0.050
DSC_0166_segment_1_segmented	DSC_0166_segment_224_segmented	val	0.448	7.126	0.247	0.130	0.517	0.048
DSC_0171_segment_1_segmented	DSC_0171_segment_224_segmented	val	0.336	0.629	0.142	0.153	0.320	0.071

Patch ID	Leaf ID	Split	Unc	AbsErr	μ^*	$\hat{\mu}$	λ^*	$\hat{\lambda}$
DSC_0183_segment_1_segmented	DSC_0183_segment_1_segmented	224	112	784	3.104	0.556	0.376	0.228	-0.017
DSC_0192_segment_1_segmented	DSC_0192_segment_1_segmented	224	112	280	6.314	0.292	0.083	0.310	0.026
DSC_0198_segment_1_segmented	DSC_0198_segment_1_segmented	224	112	324	0.164	0.600	0.101	0.350	0.110
DSC_0201_segment_1_segmented	DSC_0201_segment_1_segmented	224	112	336	1.730	0.430	0.243	0.456	0.100
DSC_0206_segment_1_segmented	DSC_0206_segment_1_segmented	224	112	268	5.430	0.050	0.376	0.155	-0.056
DSC_0217_segment_1_segmented	DSC_0217_segment_1_segmented	224	112	310	0.389	0.190	0.143	0.156	0.047
DSC_0241_segment_1_segmented	DSC_0241_segment_1_segmented	224	112	448	0.286	0.200	0.395	0.147	0.448
DSC_0251_segment_1_segmented	DSC_0251_segment_1_segmented	224	112	268	7.706	0.496	0.226	0.244	-0.072
DSC_0258_segment_1_segmented	DSC_0258_segment_1_segmented	224	112	224	0.732	0.185	0.228	0.454	0.154
DSC_0263_segment_1_segmented	DSC_0263_segment_1_segmented	224	112	224	1.161	0.600	0.415	0.579	0.401
DSC_0268_segment_1_segmented	DSC_0268_segment_1_segmented	224	112	201	5.311	0.493	0.308	0.242	0.131
DSC_0273_segment_1_segmented	DSC_0273_segment_1_segmented	224	112	280	0.302	0.106	0.352	0.206	-0.018
DSC_0278_segment_1_segmented	DSC_0278_segment_1_segmented	224	112	246	7.032	0.182	0.430	0.562	0.009
DSC_0287_segment_1_segmented	DSC_0287_segment_1_segmented	224	112	336	7.156	0.217	0.238	0.275	0.016
DSC_0288_segment_1_segmented	DSC_0288_segment_1_segmented	224	112	347	5.000	0.262	0.275	0.050	0.026
DSC_0363_segment_1_segmented	DSC_0363_segment_1_segmented	224	112	112	7.608	0.192	0.547	0.174	-0.099
DSC_0371_segment_1_segmented	DSC_0371_segment_1_segmented	224	112	134	7.796	0.372	0.531	0.299	0.110
DSC_0378_segment_1_segmented	DSC_0378_segment_1_segmented	224	112	156	7.524	0.050	0.305	0.327	0.323
DSC_0407_segment_1_segmented	DSC_0407_segment_1_segmented	224	112	268	5.405	0.436	0.245	0.582	-0.032

14.5 Casewise External 18.7 Patch Oracle Records

Table 12: Casewise patch oracle parameters on external 18.7 subset.

Patch ID	Leaf ID	Score	Sec	Calls	μ	λ	Diff	α
DSC_0003_segment_1_segmented	DSC_0003_segment_1_segmented	224	112	224	4	0.467	0.194	0.233	0.0088	0.21
DSC_0004_segment_1_segmented	DSC_0004_segment_1_segmented	224	112	448	4	0.273	0.100	0.350	0.0202	0.23
DSC_0005_segment_1_segmented	DSC_0005_segment_1_segmented	224	112	448	4	0.122	0.400	0.163	0.0132	0.20
DSC_0006_segment_1_segmented	DSC_0006_segment_1_segmented	224	112	784	4	0.152	0.325	0.151	0.0090	0.23
DSC_0007_segment_1_segmented	DSC_0007_segment_1_segmented	224	112	672	4	0.138	0.426	0.216	0.0209	0.05
DSC_0008_segment_1_segmented	DSC_0008_segment_1_segmented	224	112	448	4	0.184	0.585	0.069	0.0015	0.14
DSC_0009_segment_1_segmented	DSC_0009_segment_1_segmented	224	112	448	4	0.055	0.279	0.321	0.0299	0.20
DSC_0010_segment_1_segmented	DSC_0010_segment_1_segmented	224	112	560	4	0.330	0.452	0.127	0.0098	0.05
DSC_0011_segment_1_segmented	DSC_0011_segment_1_segmented	224	112	448	4	0.050	0.050	0.152	0.0025	0.20
DSC_0012_segment_1_segmented	DSC_0012_segment_1_segmented	224	112	448	4	0.332	0.565	0.123	0.0086	0.08
DSC_0013_segment_1_segmented	DSC_0013_segment_1_segmented	224	112	448	4	0.546	0.052	0.117	0.0205	0.05
DSC_0014_segment_1_segmented	DSC_0014_segment_1_segmented	224	112	560	4	0.564	0.498	0.192	0.0079	0.12
DSC_0015_segment_1_segmented	DSC_0015_segment_1_segmented	224	112	448	4	0.544	0.317	0.099	0.0433	0.16
DSC_0016_segment_1_segmented	DSC_0016_segment_1_segmented	224	112	448	4	0.373	0.172	0.133	0.0256	0.13
DSC_0017_segment_1_segmented	DSC_0017_segment_1_segmented	224	112	560	4	0.248	0.465	0.270	0.0112	0.06
DSC_0018_segment_1_segmented	DSC_0018_segment_1_segmented	224	112	560	4	0.600	0.523	0.163	0.0128	0.16
DSC_0019_segment_1_segmented	DSC_0019_segment_1_segmented	224	112	448	4	0.154	0.061	0.252	0.0309	0.08
DSC_0020_segment_1_segmented	DSC_0020_segment_1_segmented	224	112	448	4	0.467	0.271	0.050	0.0474	0.15
DSC_0021_segment_1_segmented	DSC_0021_segment_1_segmented	224	112	448	4	0.100	0.253	0.180	0.0259	0.07
DSC_0022_segment_1_segmented	DSC_0022_segment_1_segmented	224	112	560	4	0.575	0.253	0.084	0.0332	0.13

15 Ablation Notes and Statistical Reading of Current Run

The current run is intentionally conservative in per-leaf patch count so that all required stages could be completed end-to-end with real optimization/refinement inside one reproducible package. Under this constraint, the observed error profile is expected to be dominated by under-sampling of within-leaf heterogeneity rather than solver instability.

We summarize the practical implications:

- Patch-level MAE is currently high in full parameter space, so direct patch prediction should be treated as a routing prior rather than a final estimate.
- Refinement budgets improve score recovery behavior but do not yet fully close parameter error to patch oracle for all cases.
- Selective mixture aggregation is consistently stronger than pure direct aggregation operators, confirming the utility of uncertainty-aware routing.
- External 18.7 behavior is now measured with real oracle subset rows, so robustness statements are evidence-backed.

From a study-maturity perspective, these outcomes are sufficient for a full paper-length report because they provide real quantitative and qualitative evidence for both strengths and failure modes of the patch formulation.

16 Practical Next Experiments from Current Evidence

The current run establishes a working real patch stack and exposes specific optimization priorities:

- Increase patches per leaf in oracle generation to better represent intra-leaf variability.
- Replace current patch features with stronger pretrained patch encoders while preserving leakage-safe parent-leaf splits.
- Refit routing thresholds on larger validation subsets and report confidence intervals for route proportions.
- Re-run aggregation experiments with uncertainty calibration improvements and domain-adaptive weighting.
- Expand external 18.7 oracle subset beyond 20 patches to reduce external metric variance.

These recommendations are directly grounded in observed non-empty outputs rather than planned-only infrastructure.

A Additional Notes

Future versions should increase patches per leaf, include stronger image encoders, and raise oracle/refinement budgets under SLURM for a higher-fidelity patch model.